



Private Pension Plan

Case 3 A – Group 3
Financial Engineering

Presentation Roadmap



- 1 Client Profile & Needs
- 2 Macro & Market Context
- 3 German Statutory Pension Analysis
- 4 Gap Analysis & Scenario Drivers
- 5 Investment Instrument Research
- 6 Investment Strategy Design
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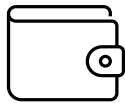
Client Introduction



- Professor, 45 years old (private university, Germany)



- Married, wife 43 (not working)
- Two children (5 & 8)
- Own house, no mortgage
- Private health insurance (€700/month)



- Current income: €81,600 gross/year → €57,000 net/year
- Plans to retire at 68 (2048), no other private pension plan

► Starting Point

- Lump sum: €250,000 to invest
- Statutory pension expected: €46k/year gross (€25k net) at retirement
- Pension gap: **€26k**/year net (today's euros)

Objectives



Goal 1

Generate €26k/year
(today's euros at
retirement)



Goal 2

Maintain lifestyle
(80% of today's net
income)



Goal 3

Balance growth &
downside protection
(risk averse)



Goal 4

No liquidity need
before retirement

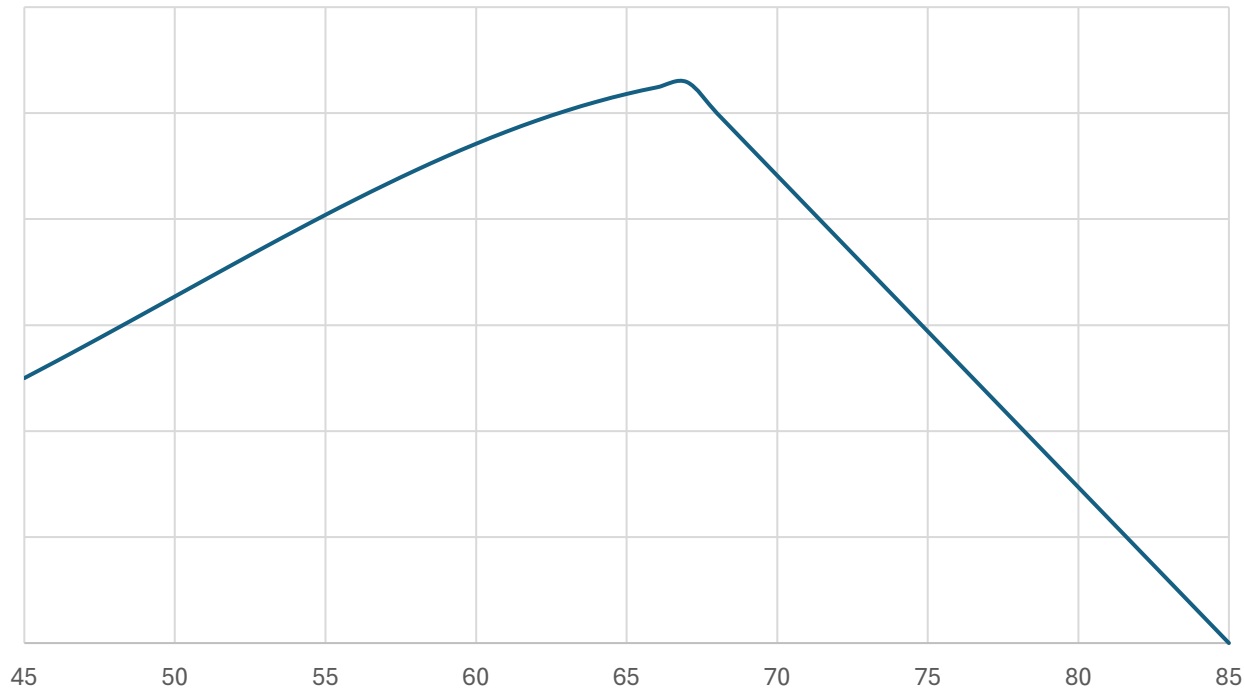


Goal 5

Optimize for net-of-
tax outcomes

Investment Requirement & Major Assumptions

Private Pension Balance Development



EUR-based liabilities → minimize FX risk (EUR bonds, some unhedged USD equities acceptable but should avoid).



German pension/tax system applies.

Investment Horizon 22 years (from age 45 to 67, long-term oriented)

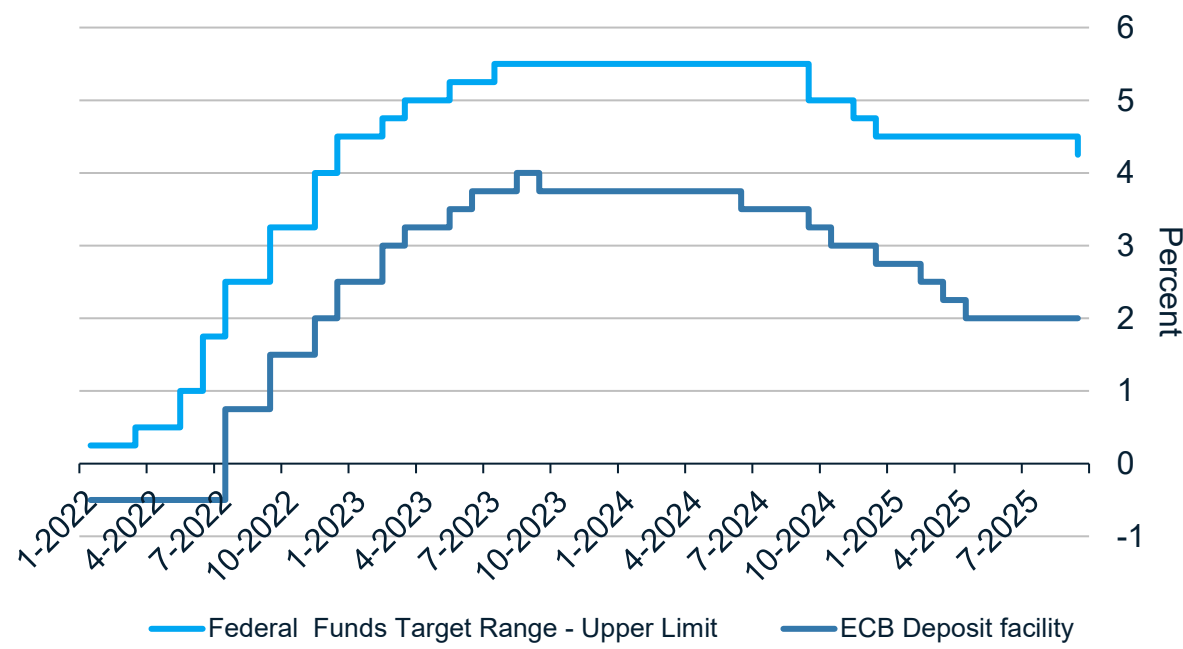
Liquidity No interim withdrawals, capital locked until retirement

Return Target 6–7% nominal (~4% real) to close pension gap.

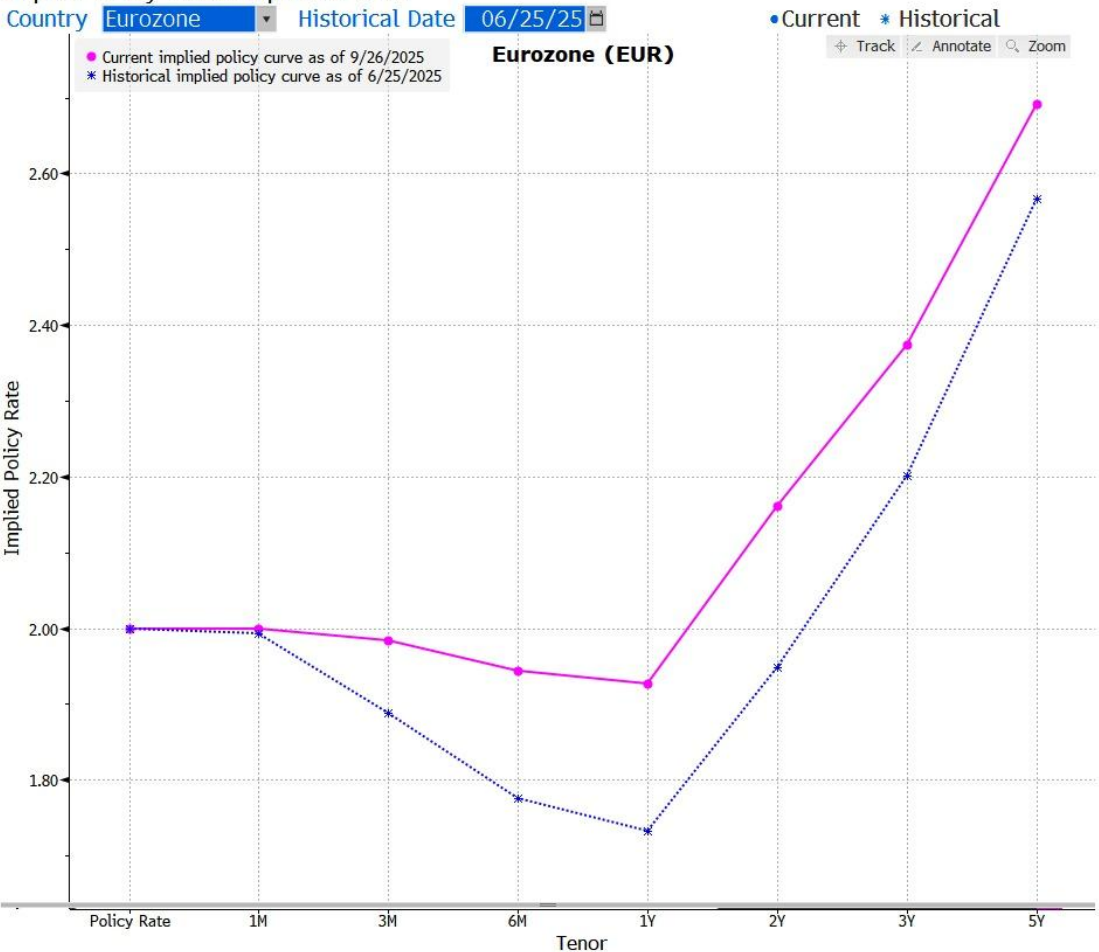
Risk Profile Moderate / slightly risk-averse.

Current Interest Rate Environment

Central Bank Rates From 2022 to date



Implied Policy Rate Comparison Chart



- Higher for Longer Rate Environment with inflation upside risks and geo-political uncertainty.

Macro Outlook

Over the next medium term, economic growth in the **Euro Area** is projected to moderate to 1.3% in 2027. Inflation (HICP) is expected to decline from ~2.1% to ~1.7% and then settle near ~1.9%. Meanwhile, in the **United States**, real GDP growth is expected to stay under 2%, Core PCE inflation hovering around 2–2.4%. Globally, the IMF estimates growth of ~3.0–3.3% in 2025, with headline inflation at ~4.2% in 2025 gradually easing in 2026.

Growth and inflation projections for the euro area

(annual percentage changes, revisions in percentage points)

	September 2025				Revisions vs June 2025			
	2024	2025	2026	2027	2024	2025	2026	2027
Real GDP	0.8	1.2	1.0	1.3	0.0	0.3	-0.1	0.0
HICP	2.4	2.1	1.7	1.9	0.0	0.1	0.1	-0.1
HICP excluding energy and food	2.8	2.4	1.9	1.8	0.0	0.0	0.0	-0.1

German Statutory Pension System

Financing

Employer + Employee each contribute **50%: 9.3%** of gross salary, total contribution **18.6%**.

Benefit Determination

Pension Points = Individual Earnings ÷ German Average Earnings
(Rentenpunkte) (Durchschnittsentgelt)

Retirement Pension = Total Points × Current Pension Value
(Bruttorente) (Aktueller Rentenwert)

Simple Example

In 2025, 45 years old

Annual Gross Earning = €81,600

German Average Earnings = €50,493

Pension Points Earned in this year = 1.62 points



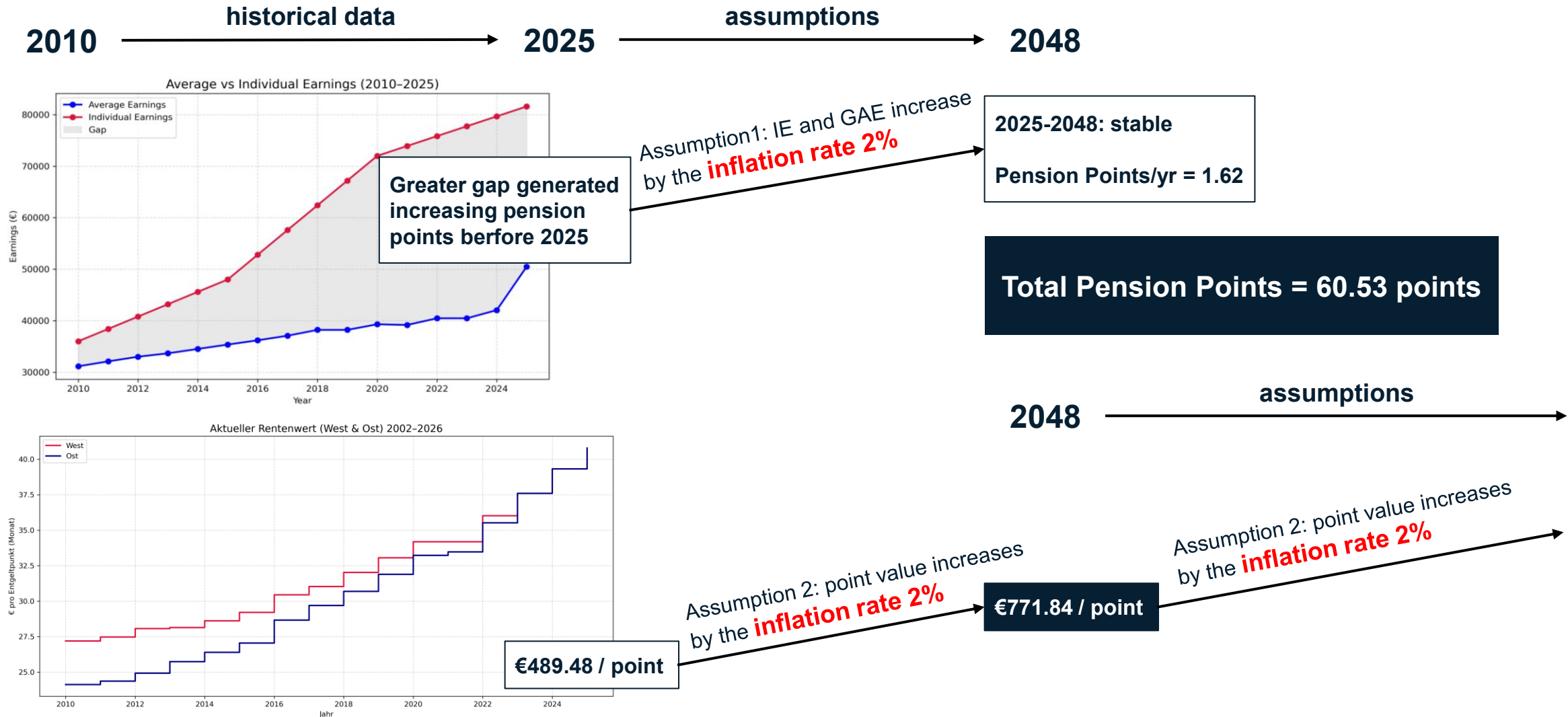
In 2048, 68 years old

Cumulative Pension Points for 38 years = 60.53 points

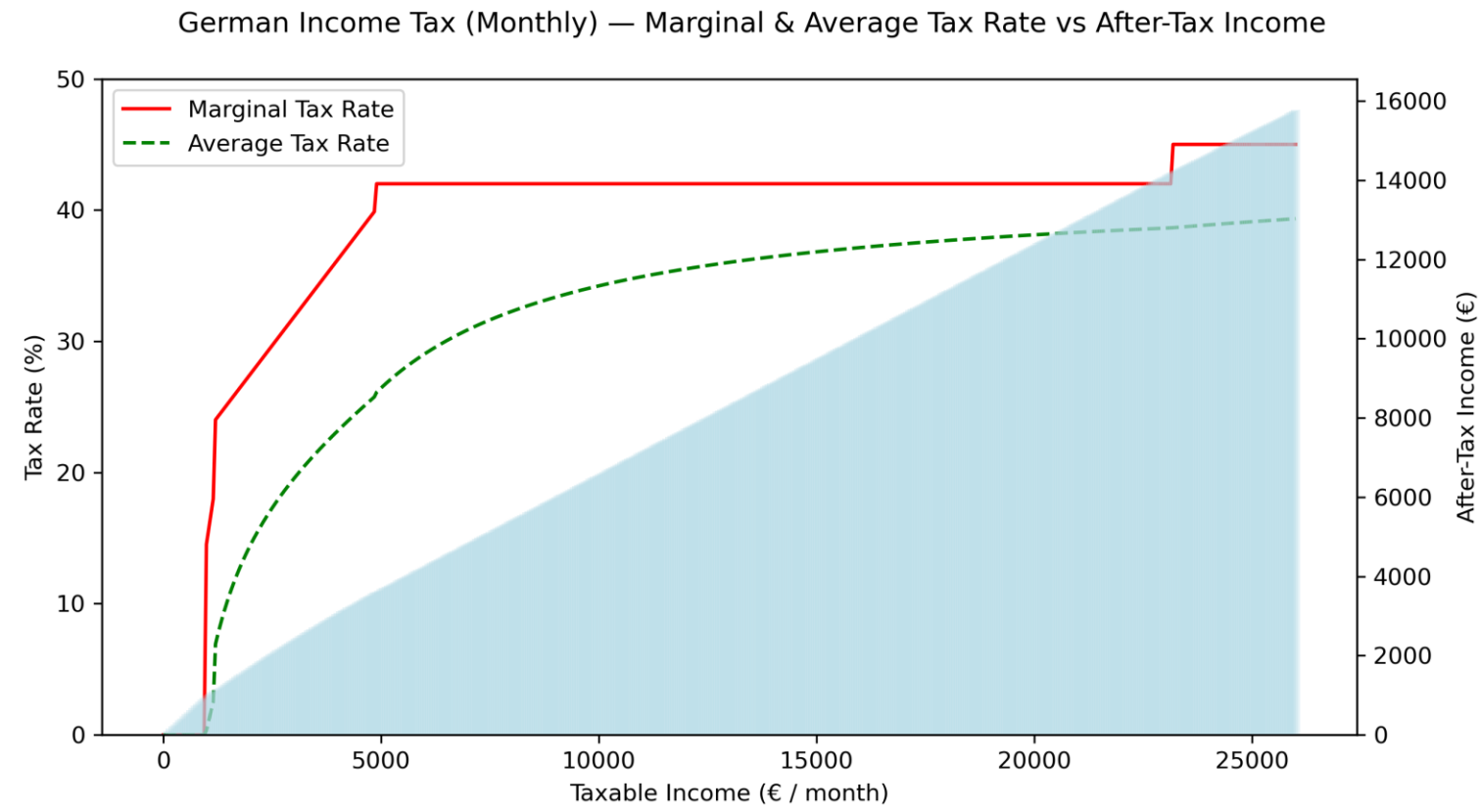
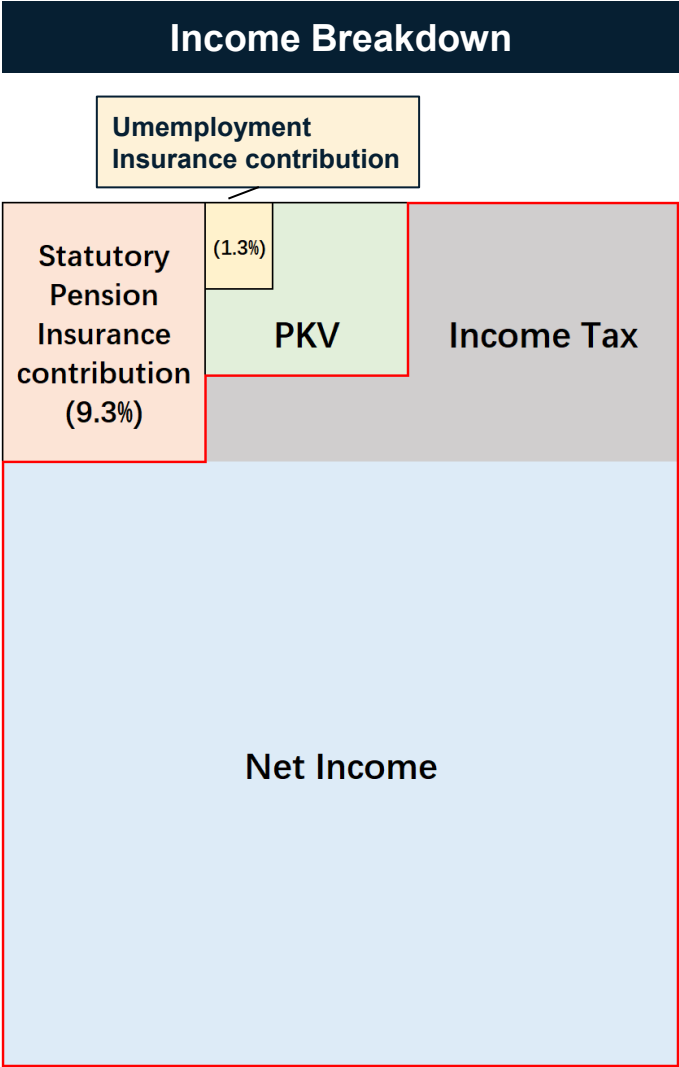
Annual Current Pension Value = €771.84 / point
(Aktueller Rentenwert)

Retirement Pension = €45,808
(Bruttorente)

Macro Assumptions



From Gross to Net: Income



In 2047, 67 years old, last working year:
Annual Net Income = €84509

From Gross to Net: Statutory Pension

Step 1: Determine Gross Pension

Step 2: Apply Taxable Portion

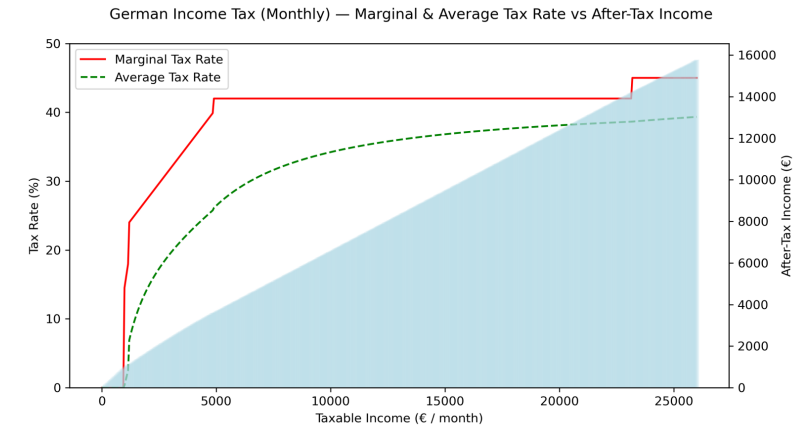
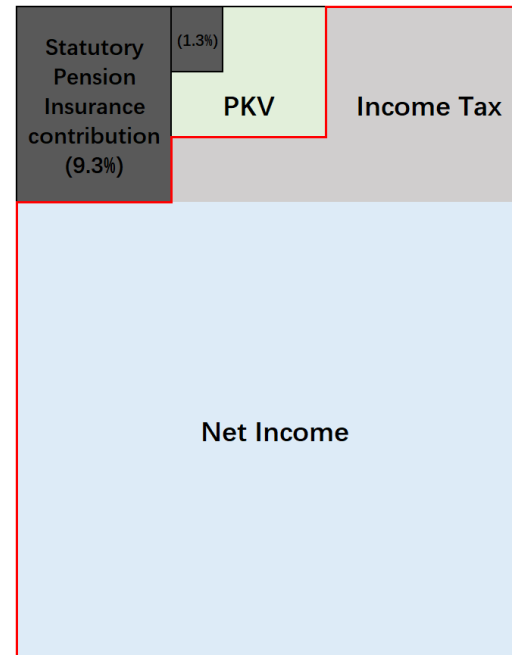
„Ab 2023 steigt der Besteuerungsanteil nur noch um einen halben Prozentpunkt pro Jahr an. Damit wird die vollständige Besteuerung der Renten (100 %) erst im Jahr 2058 erreicht.“

Taxable Pension Income = Gross Pension × 95.5%

Step 3: Deduct Allowable Expenses

- ☐ Health insurance contributions
- ☐ Long-term care insurance contributions
- ☐ Other deductible provisions

Step 4: Calculate Taxable Income(zvE) Tax



Gap between Needed Cashflow and Statutory Pension

Retirement Income Replacement Rate: the proportion of pre-retirement income that is typically needed in retirement to maintain the same living standard.

“A common rule of thumb is that total retirement income—Social Security plus pensions, asset income, and other sources—should replace about 70 percent of preretirement earnings.”

— SSA – Alternate Measures of Replacement Rates (Policy Paper)

Needed Cashflow in 2048

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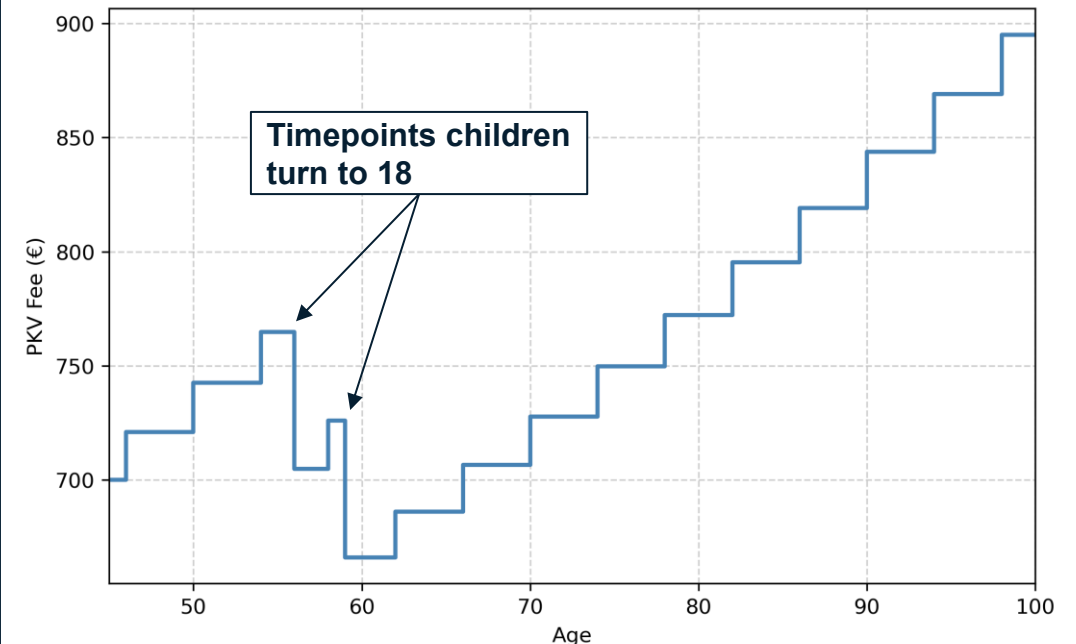
80% × Last Net Income

+

Full Private Health Insurance Fee

Assumptions:

- PKV fee increases every **3 years**, increase rate **3%**
- Child's PKV fee = **€60/month**



Scenario Insights – What Drives the Pension Gap?

Scenario Summary						
	Current Values:	Scenario 1_Inflation Increase	Scenario2_Inflation3%	Scenario3_Inf_HI_Increase	All_Increase	
Changing Cells:						
\$Q\$2	2%	3%	3%	3%	3%	
\$Q\$3	3%	3%	3%	4%	5%	
\$Q\$7	80%	80%	80%	80%	85%	
\$Q\$8	2%	2%	2%	2%	2%	
\$Q\$9	2%	2%	2%	2%	2%	
\$Q\$10	2%	2%	2%	2%	2%	
Result Cells:						
Pension Gap	25,390	32,293	39,697	32,064	44,425	

- Raising inflation to 3% widens the gap to €32–40k/year (+25–55%).
- Pension sustainability is highly sensitive to price stability.
- Increasing replacement rate from 80% → 90% pushes the gap to ~€34k/year (+34%).
- Higher lifestyle expectations in retirement materially raise the funding need.
- Premium growth of 5% vs 3% adds ~€6k to the gap.
- Medical inflation is a risk to watch, though less dominant than general inflation.

Scenario Insights – What Drives the Pension Gap?

Scenario Summary					
	Pension Credit Increase until 2050	1% Pension Credit Increase	No Increase in Pension Credit	High Replacement rate	Combined
Changing Cells:					
\$Q\$2	2%	2%	2%	2%	3%
\$Q\$3	3%	3%	3%	3%	4%
\$Q\$7	80%	80%	80%	90%	85%
\$Q\$8	2%	1%	0%	2%	2%
\$Q\$9	1%	1%	0%	2%	0%
\$Q\$10	0%	0%	0%	2%	0%
Result Cells:					
Pension Gap	28,241	33,029	40,035	34,010	42,304

Pension credit indexation matters for downside protection.

- If pension credits are not increased, the gap remains above €40k.
- Indexation protects against severe downside.

Worst-case scenario is severe.

- Combining high inflation, high healthcare cost growth, and higher replacement rate pushes the gap to ~€42k — almost 70% higher than the base case (€25k).

Required returns rise to 7–8%, which is much harder to achieve sustainably. Portfolio must be designed to deliver 5–6% nominal CAGR in base case

Why ETFs for the Pension Portfolio?

- **Diversification** → one ETF = hundreds/thousands of stocks or bonds. Reduces single-company/issuer risk
- **Cost Efficiency** → very low management fees (TER ~0.1–0.3%) compared to active funds
- **Transparency** → ETFs track clear indices (MSCI World, Euro Gov Bonds, etc.), easy to monitor
- **Liquidity & Accessibility** → traded daily on exchanges like stocks; easy to buy/sell for private investors
- **Tax Efficiency** → equity ETFs enjoy 30% partial exemption (Teilfreistellung) in Germany
- **Global Access** → exposure to markets (Emerging, US, REITs, gold) otherwise hard for individual investors to replicate

ISHARES CORE MSCI WORLD

Security Information

- ISIN: IE00B4L5Y983
- Fund Type: ETF
- Asset Group: Global Equity
- Category Group: Equity Developed World
- Geo. Focus: Global
- Expense Ratio: 0.20%
- Index Weight: Market Cap
- Rebalancing Frequency: Quarterly

Top Industry Group

- Internet (10.88%)
- Semiconductor (10.39%)
- Software (8.93%)
- Banks (7.89%)

Top Geographic

- U.S. (70.67%)
- Japan (5.45%)
- U.K. (3.52%)
- Canada (3.29%)
- France (2.58%)
- Switzerland (2.57%)

Metrics

- Annual Return: 13.11%
- Volatility: 15.7%
- Sharpe Ratio: 0.66

Security Description

iShares Core MSCI World UCITS ETF is an exchange-traded fund incorporated in Ireland. The Fund aims to track the performance of the MSCI World Index.

ISHARES CORE EM IMI ACC

Security Information

- ISIN: IE00BKM4GZ66
- Fund Type: ETF
- Asset Group: Global Equity
- Category Group: Equity Emerging Markets Global
- Geo. Focus: Emerging Markets
- Expense Ratio: 0.180%
- Rebalancing Frequency: Quarterly

Top Industry Group

- Semiconductors (16.08%)
- Banks (13.41%)
- Internet (12.23%)
- Diversified Finan Serv (4.64%)

Top Geographic

- China (21.44%)
- Taiwan (19.76%)
- India (16.81%)
- South Korea (11.46%)
- Hong Kong (4.45%)
- Brazil (4.25%)

Metrics

- Annual Return: 9.68%
- Volatility: 18.72%
- Sharpe Ratio: 0.37

Security Description

iShares Core MSCI EM IMI UCITS ETF is an exchange-traded fund incorporated in Ireland. The Fund aims to track the performance of the MSCI Emerging Markets IMI Index.

ISHARES GLBL INFRASTRUCTURE

Security Information

- ISIN: IE00B1FZS467
- Fund Type: ETF
- Asset Group: Sector Equity
- Category Group: Equity Thematic
- Expense Ratio: 0.650%
- Index Weight: Market Cap
- Rebalancing Frequency: Quarterly

Top Industry Group

- Electric (46.81%)
- Transportation (15.95%)
- Pipelines (15.73%)
- REITS (5.28%)

Top Geographic

- U.S. (63.03%)
- Canada (12.60%)
- Japan (4.28%)
- U.K. (3.69%)
- India (2.09%)
- Hong Kong (1.92%)

Metrics

- Annual Return: 9.13%
- Volatility: 14.33%
- Sharpe Ratio: 0.44

Security Description

iShares Global Infrastructure UCITS ETF is an exchange-traded fund incorporated in Ireland. The Fund aims to track the performance of FTSE Global Core Infrastructure Index.

ISHARES CORE EURO GOVT BOND

Security Information

- ISIN: IE00B4WXJJ64
- Fund Type: ETF
- Asset Group: Global Fixed Income
- Category Group: Global Government Bond
- Geo. Focus: Eurozone
- Expense Ratio: 0.07%
- Index Weight: Market Cap
- Rebalancing Frequency: Quarterly

Top Industry Group

- Sovereign (99.86%)
- Banks (0.01%)

Top Geographic

- France (23.95%)
- Italy (21.91%)
- Germany (18.80%)
- Spain (14.31%)
- Belgium (4.91%)
- Netherlands (4.04%)

Metrics

- Annual Return: 1.83%
- Volatility: 4.93%
- Sharpe Ratio: -0.19 (Irrelevant)

Security Description

iShares Core EUR Govt Bond UCITS ETF is an exchange-traded fund incorporated in Ireland. The Fund aims to track the performance of the Bloomberg Euro Treasury Bond Index.

ISHARES CORE EURO CORP BOND

Security Information

- ISIN: IE00B3F81R35
- Fund Type: ETF
- Asset Group: Global Fixed Income
- Category Group: Global Corporate Bond
- Rating: Investment Grade BBB or higher
- Expense Ratio: 0.09%
- Index Weight: Market Cap
- Rebalancing Frequency: Monthly

Top Industry Group

- Banks (31.51%)
- Electric (5.72%)
- Telecommunications (4.92%)
- Auto Manufacturers (4.82%)

Top Geographic

- France (19.73%)
- U.S. (15.76%)
- Netherlands (15.40%)
- Germany (8.46%)
- U.K. (6.96%)
- Spain (5.88%)

Metrics

- Annual Return: 3.84%
- Volatility: 3.97%
- Sharpe Ratio: 0.27

Security Description

iShares Core EUR Corp Bond UCITS ETF is an exchange-traded fund incorporated in Ireland. The Fund aims to track the performance of the Bloomberg Euro Corporate Bond Index.

ISH EB.REXX GVTGRMNY10.5 DE

Security Information

- ISIN: DE000AoD8Q31
- Fund Type: ETF
- Asset Group: Global Fixed Income
- Category Group: Fixed Income European Region
- Rating: Investment Grade BBB or higher
- Expense Ratio: 0.160%
- Rebalancing Frequency: Monthly

Top Industry Group

- Sovereign (99.96%)

Top Geographic

- Germany (99.96%)

Metrics

- Annual Return: 2.1%
- Volatility: 11.15%
- Sharpe Ratio: -0.06 (Irrelevant)

Security Description

iShares eb.rexx Government Germany 10.5+yr UCITS ETF (DE) is an open-end, UCITS compliant exchange traded fund incorporated in Germany. The fund aims to track the performance of the eb.rexx Government Germany 10.5+ index. The fund distributes income received to shareholders.

INVESCO PHYSICAL GOLD ETC

Security Information

- ISIN: IE00B579F325
- Fund Type: ETC
- Asset Group: Commodity
- Category Group: Precious Metals
- Expense Ratio: 0.120%
- Index Weight: Single Asset

Top Industry Group

- NA

Top Geographic

- NA

Metrics

- Annual Return: 12.80%
- Volatility: 14.64%
- Sharpe Ratio: 0.69

Security Description

Invesco Physical Gold ETC is a UCITS eligible Exchange Traded Certificate incorporated in Ireland. It aims to provide the performance of the spot gold price (London PM fixing price). Each Gold ETC is a certificate which is secured by physical gold held in JP Morgan Chase Bank's London vaults. The vast majority of gold is held in allocated gold bars.

ISHARES DVL MKT PROPERTY YLD

Security Information

- ISIN: IE00B1FZS350
- Fund Type: ETF
- Asset Group: Real Estate
- Category Group: Indirect Real Estate
- Expense Ratio: 0.590%
- Index Weight: Market Cap
- Rebalancing Frequency: Quarterly

Top Industry Group

- REITS (87.50%)
- Real Estate (11.06%)

Top Geographic

- U.S. (63.51%)
- Japan (8.33%)
- Australia (4.52%)
- U.K. (3.70%)
- Hong Kong (3.42%)
- Singapore (3.37%)

Metrics

- Annual Return: 11.94%
- Volatility: 17.88%
- Sharpe Ratio: 0.51

Security Description

iShares Developed Markets Property Yield UCITS ETF is an exchange-traded fund incorporated in Ireland. The Fund aims to track the performance the FTSE EPRA/NAREIT Developed Dividend+ Index.

SPDR GLOBAL REAL ESTATE

Security Information

- ISIN: IE00B8GF1M35
- Fund Type: ETF
- Asset Group: Real Estate
- Category Group: Indirect Real Estate
- Expense Ratio: 0.40%
- Index Weight: Market Cap
- Rebalancing Frequency: Quarterly

Top Industry Group

- REITS (91.22%)
- Real Estate (7.53%)

Top Geographic

- U.S. (70.10%)
- Japan (8.85%)
- U.K. (3.64%)
- Singapore (2.77%)
- Australia (2.74%)
- France (1.96%)

Metrics

- Annual Return: 7.46%
- Volatility: 16.86%
- Sharpe Ratio: 0.28

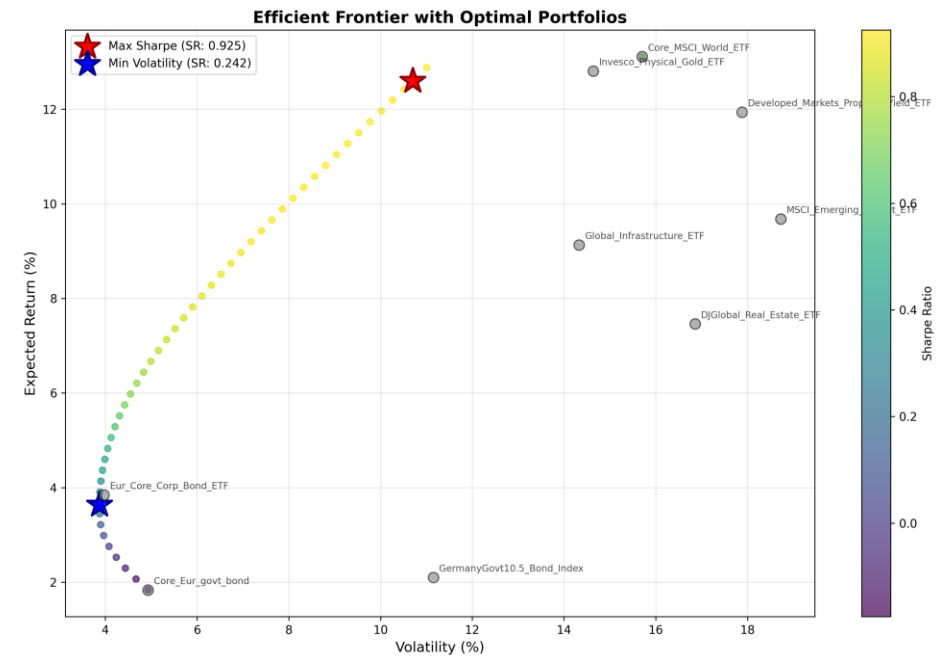
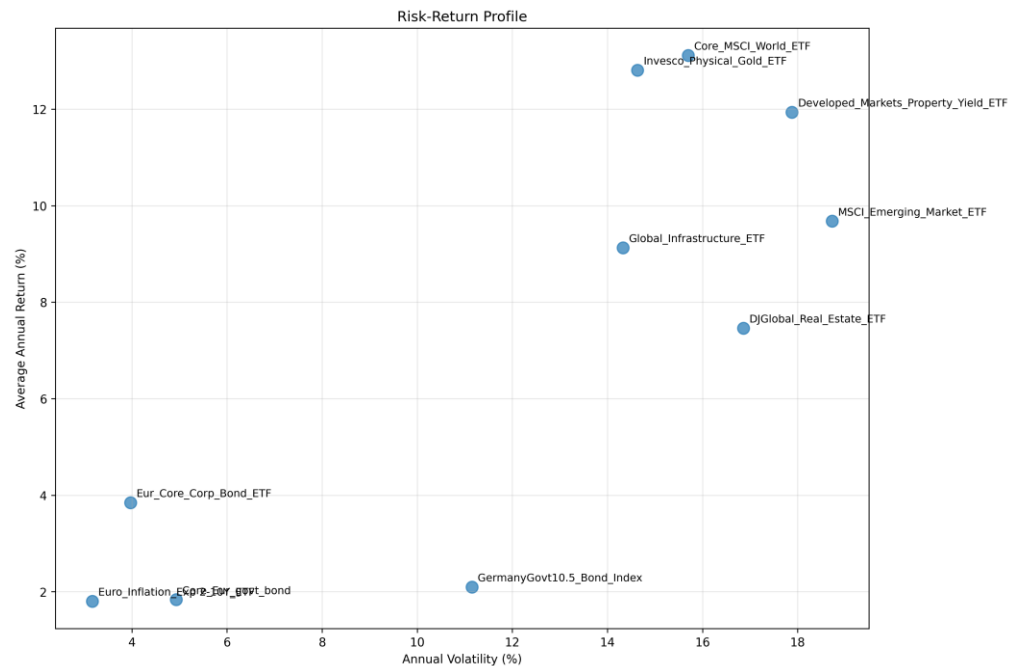
Security Description

SPDR Dow Jones Global Real Estate UCITS ETF is an open ended, UCITS compliant, physically-backed exchange traded fund, domiciled in Ireland. The fund's objective is to track the performance of the Global Real Estate Securities Market.

Efficient Portfolio

Markowitz Theory & Efficient Frontier

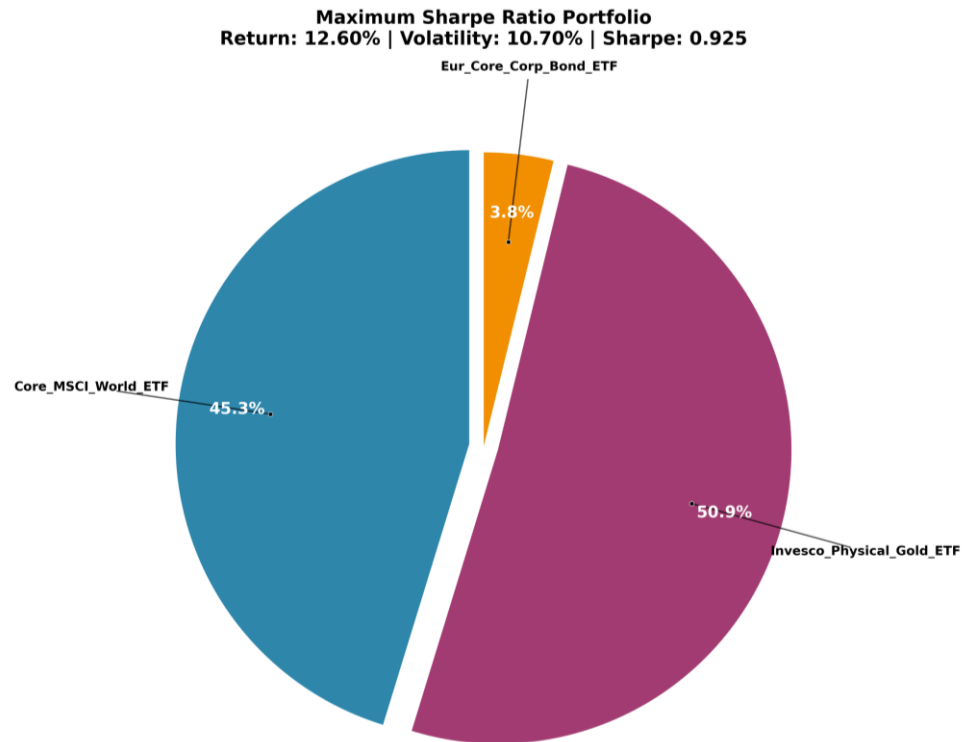
- Efficient Frontier is Curve showing optimal portfolios with maximum return for each risk level
- Diversification reduces risk without sacrificing return



Efficient Portfolio

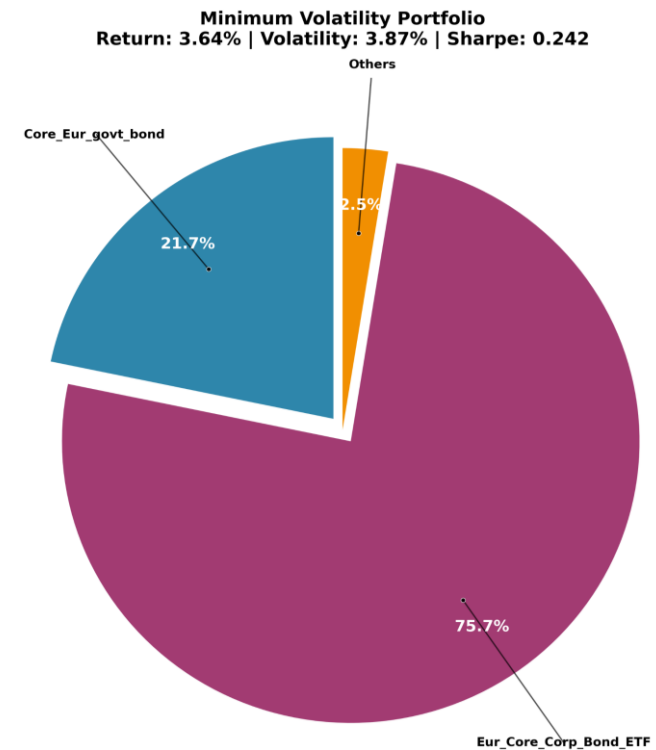
Maximum Sharpe Ratio

Best risk-adjusted return
Good For Long-term growth
For us, Too equity-heavy, exceeds risk tolerance



Minimum Volatility

Lowest possible risk, Minimise portfolio volatility
Good for Capital preservation (approaching retirement)
For us, bond-heavy, return potential too low.



Proposed Portfolio Composition

- **Core MSCI World ETF – 40%**
- **Core Euro Government Bond ETF – 20%**
- **Euro Core Corporate Bond ETF – 20%**
- **Global Infrastructure ETF – 10%**
- **Invesco Physical Gold ETF – 10%**
- Other ETFs (EM, Property, Real Estate, German Gov Bonds) – 0%

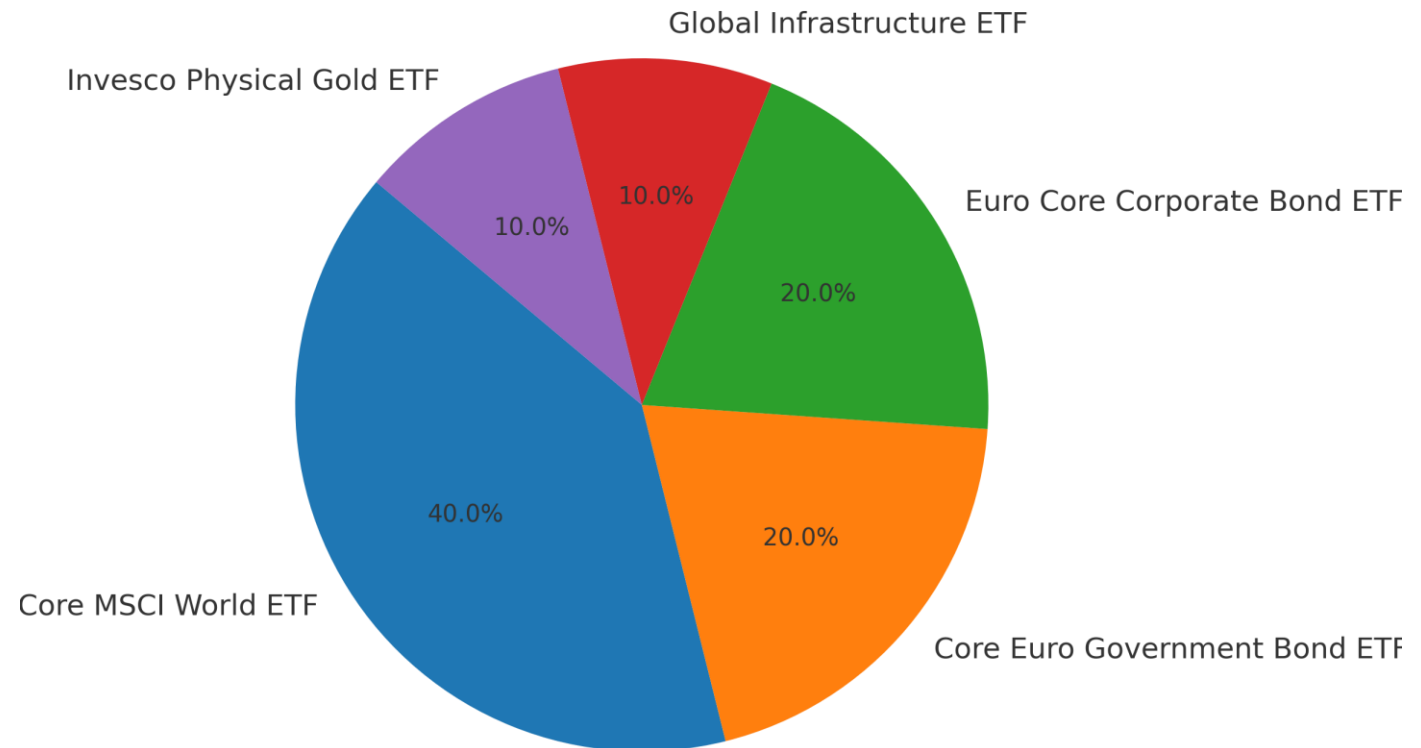
Target Outcome

Growth via global equities + infrastructure.

Stability via bonds.

Diversification & inflation hedge via gold.

Proposed Portfolio Allocation

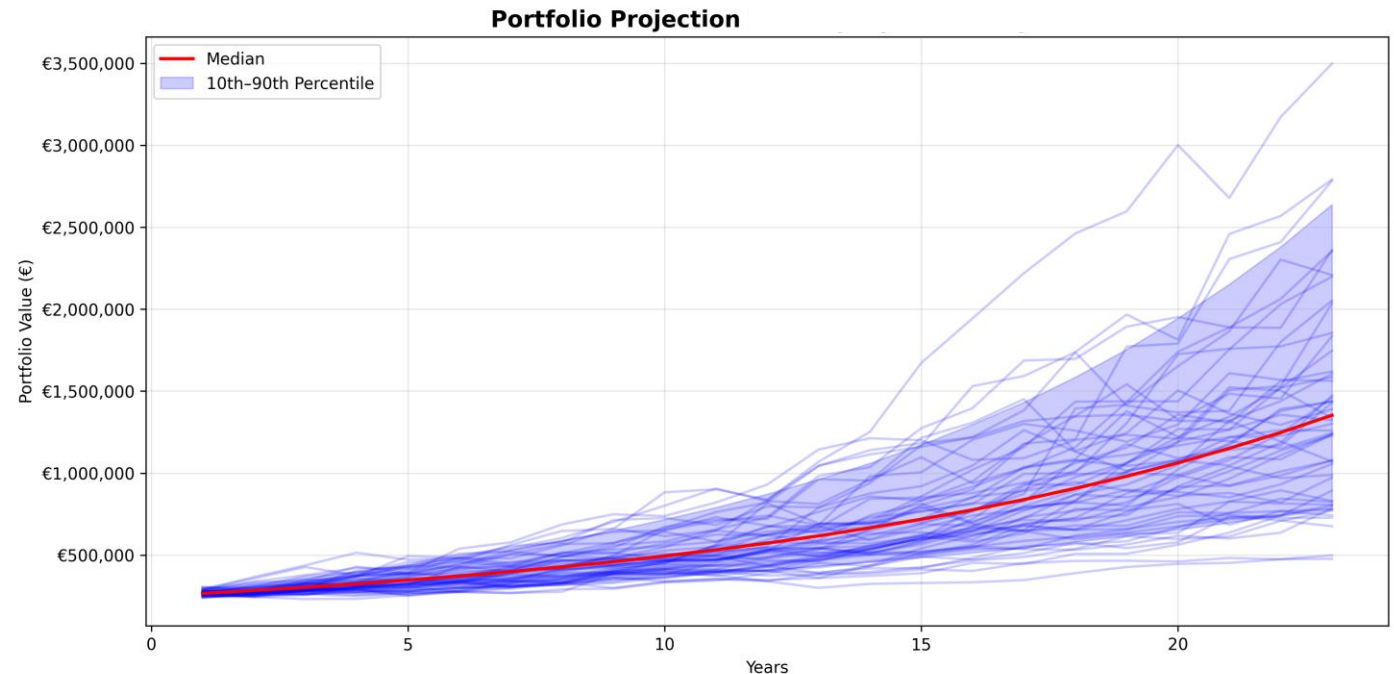


Portfolio Modelling

Methodology

- Simulates portfolio evolution using **daily i.i.d. normal returns** for each ETF individually.
- Historical data used to estimate μ (**expected return**) and σ (**volatility**) per asset.
- **100,000 Monte Carlo paths** over the investment horizon (for simplicity)

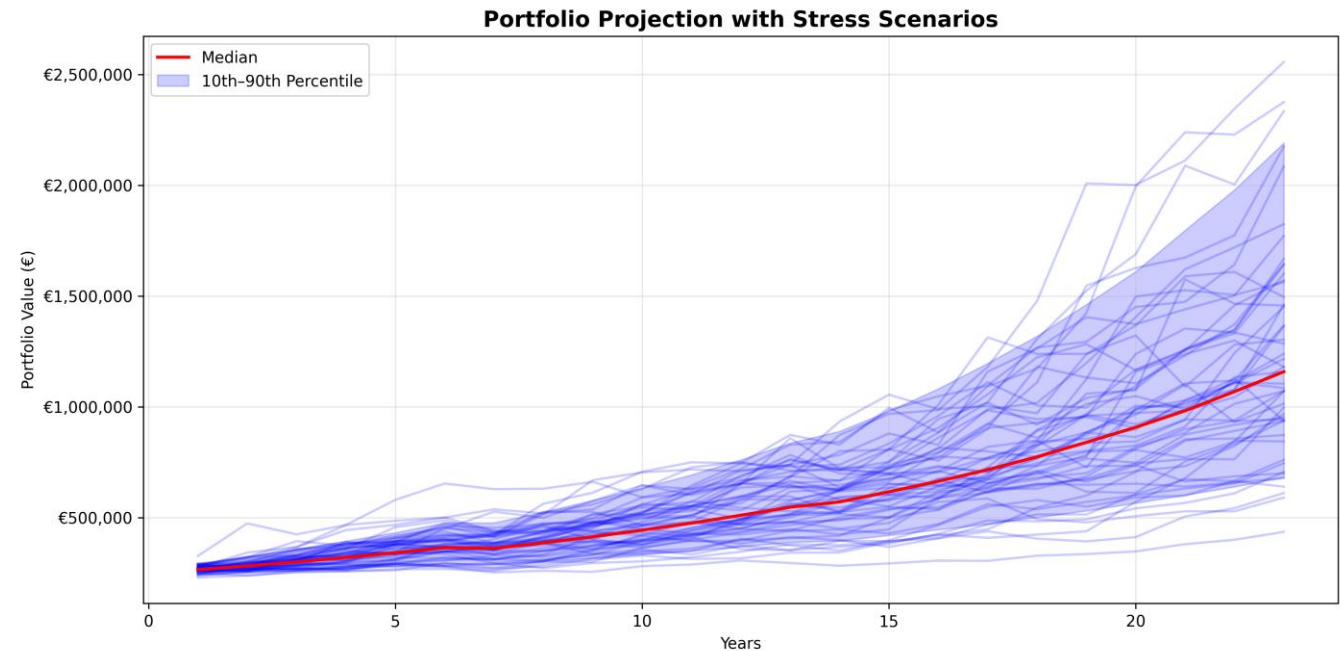
Metric	Value
Initial Value	€250,000
Median Final Value	€1,353,098
Median CAGR	7.62%
Mean CAGR	7.83%
Annualized Volatility	9.71%
Sharpe Ratio (rf=2.77%)	0.52
Annual VaR (95%)	-6.22%
Annual CVaR (95%)	-9.78%



Stress Modelling – Case 1

- We need to model risks due to external shocks like trade frictions or equity crash
- Assumes that the market is under stress by introducing a stress factor that drags the mean return down.
- Historical data used to estimate μ (**expected return**) is stressed and portfolio is then modelled similar to base case.

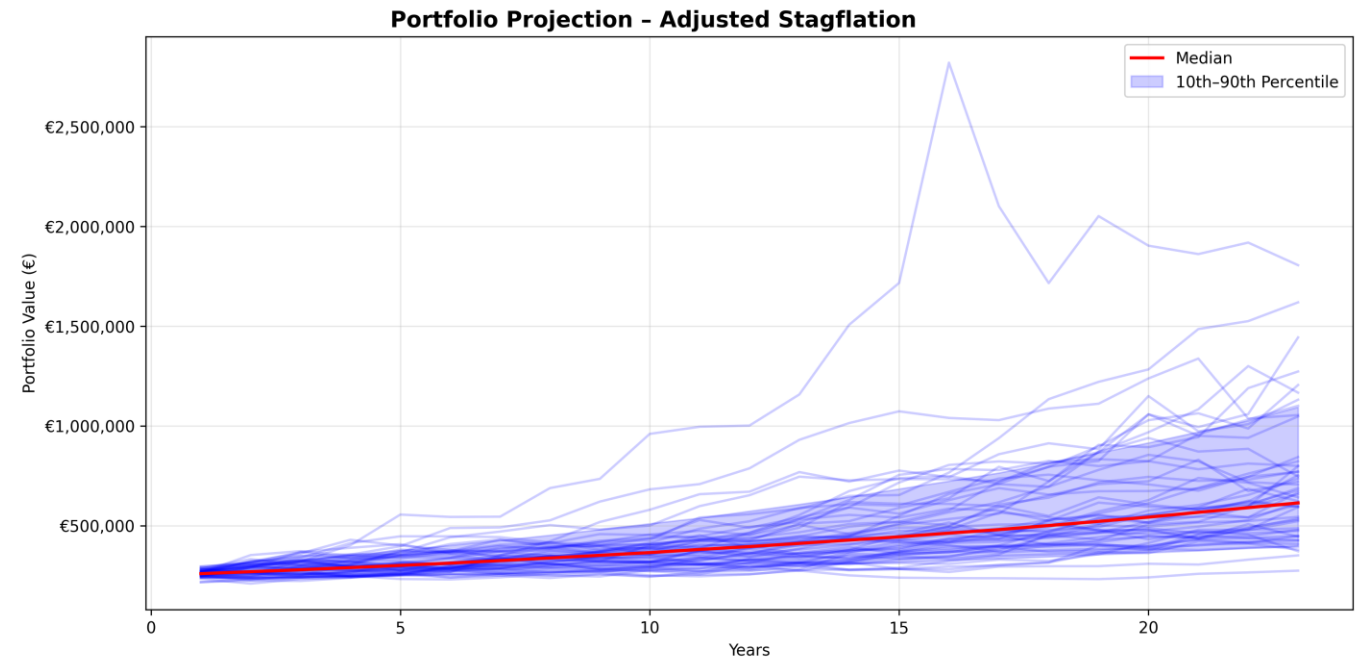
Metric	Value
Initial Value	€250,000
Median Final Value	€662,638
Median CAGR	4.33%
Mean CAGR	4.47%
Annualized Volatility	8.41%
Sharpe Ratio (rf=2.77%)	0.20
Annual VaR (95%)	-7.90%
Annual CVaR (95%)	-11.02%



Stress Modelling – Case 2

- Stagflation scenario when there is little or no growth in economy
- Assumes that the returns for equities, bonds and gold all decrease.
- Historical data used to estimate σ (**volatility**) and portfolio is then modelled similar to base case with reduced return assumption.

Metric	Value
Initial Value	€250,000
Median Final Value	€614,020
Median CAGR	3.98%
Mean CAGR	4.20%
Annualized Volatility	8.33%
Annual VaR (95%)	-8.04%
Annual CVaR (95%)	-11.22%



Rebalancing Modeling Limitations

Complexity of Assumptions

- Choice of **frequency** (monthly, quarterly, annually).
- **Transaction costs** and bid–ask spreads.
- **Tax impacts** (e.g. realization of gains in Germany).
- Strong **path dependency** on market timing.

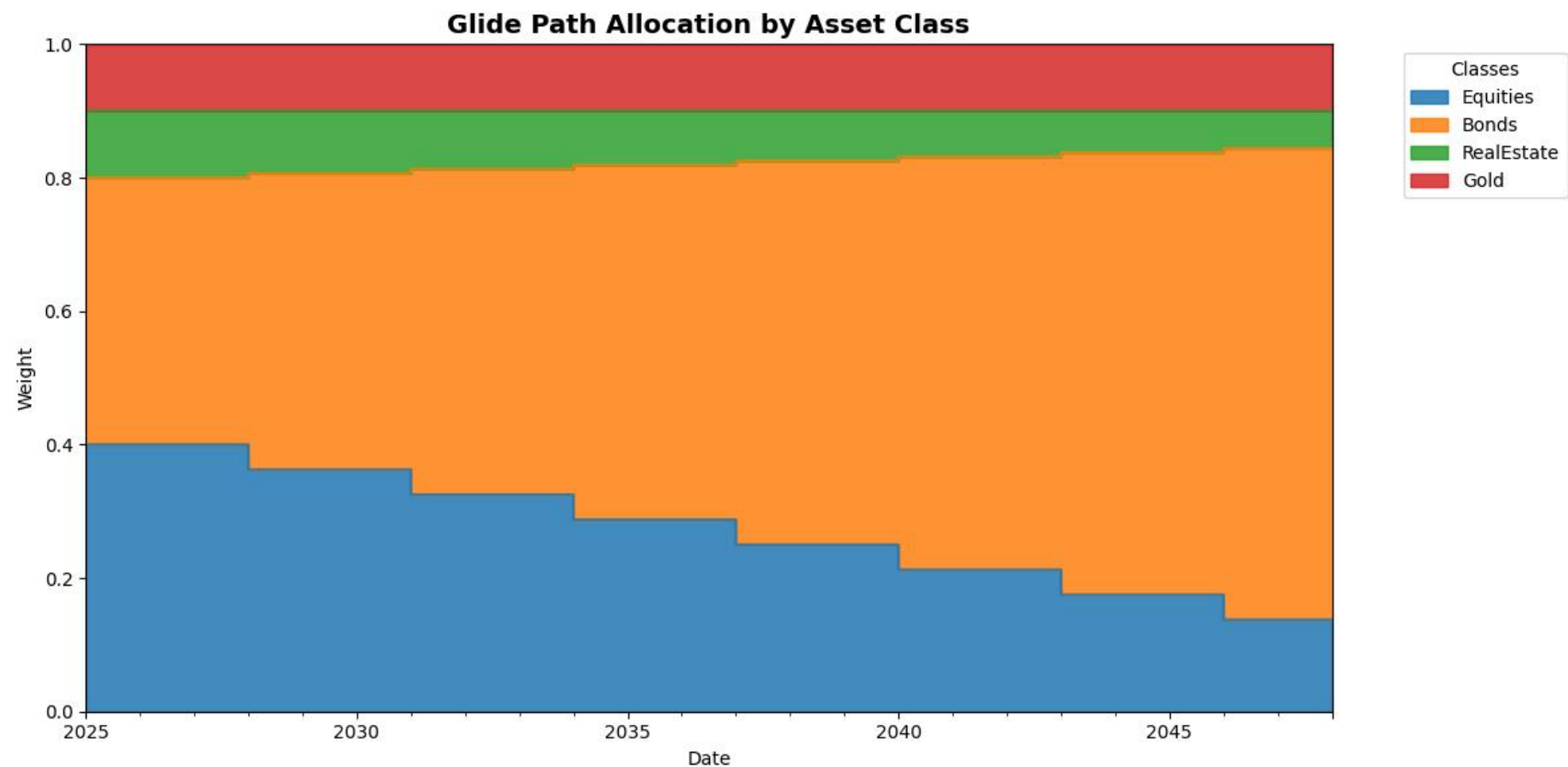
Modeling Limitations

- Requires simulating **dynamic allocation rules** vs. static weights.
- Adds heavy **computational complexity** on top of return projection.
- Realistic costs and frictions are **hard to calibrate**.

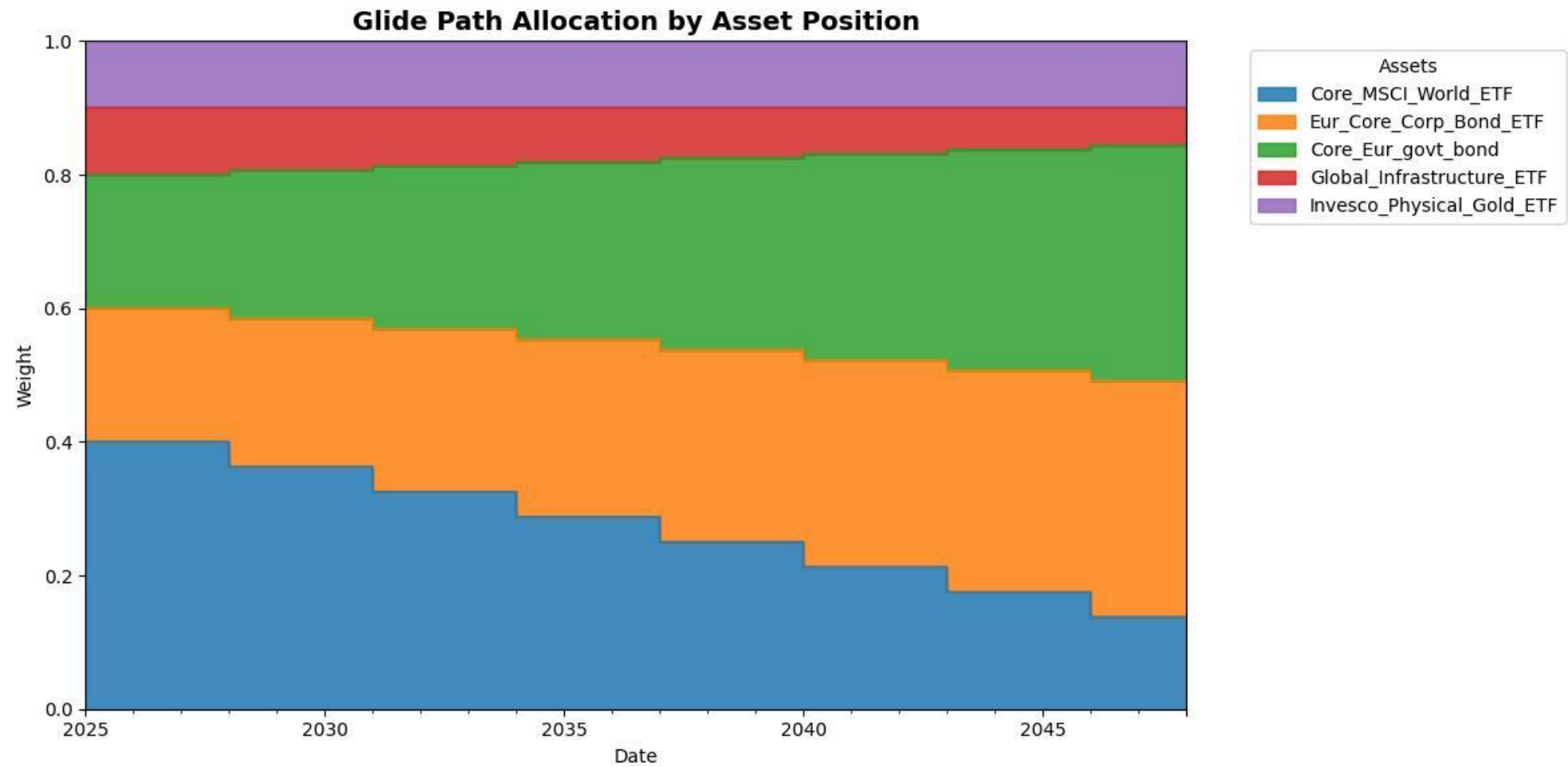
Theoretical Treatment Instead

- Rebalancing = essential for **risk control** (maintains target allocation).
- Prevents **equity drift** from dominating long-term risk profile.

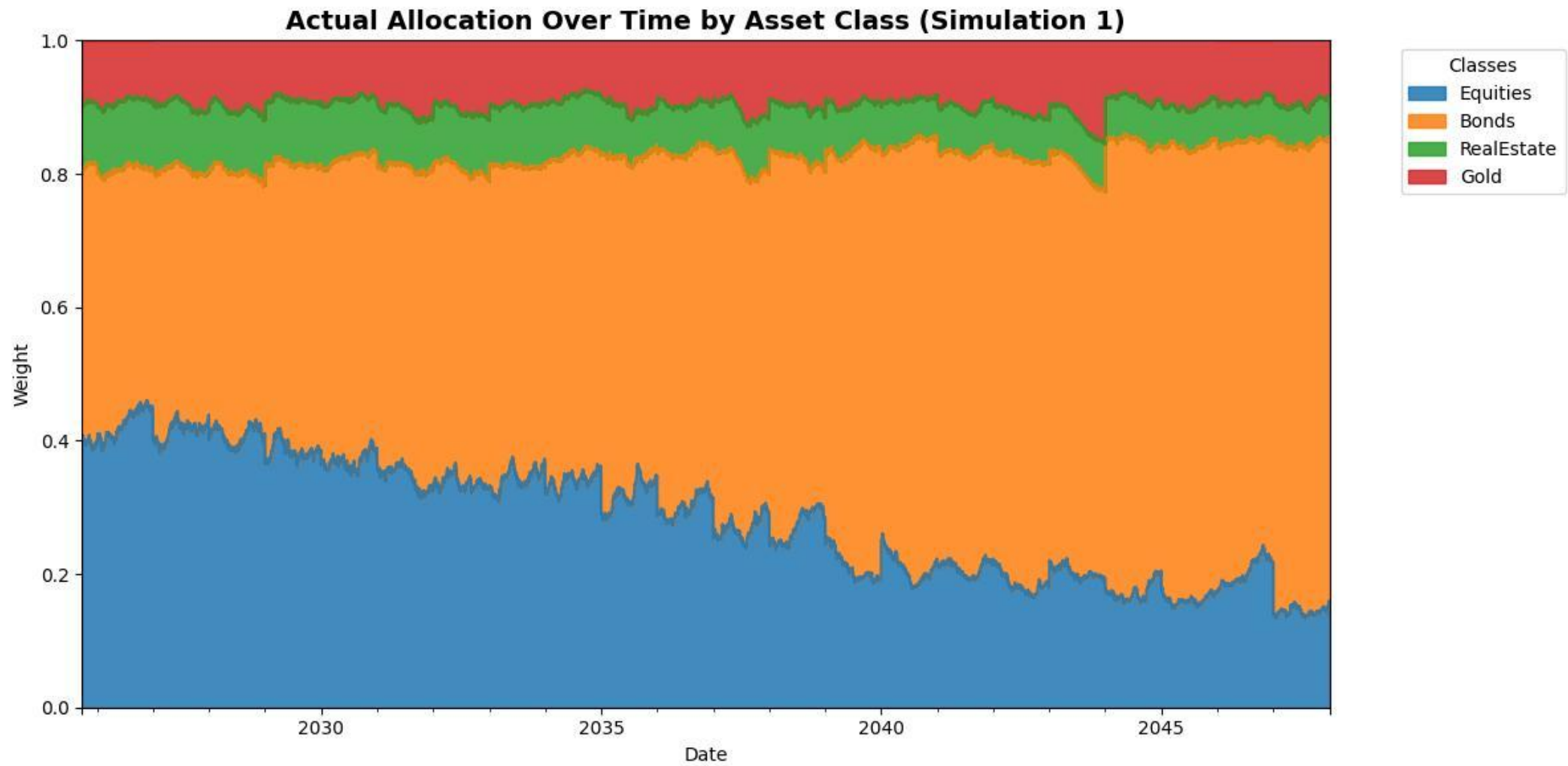
Glide Path Allocation by Asset Class



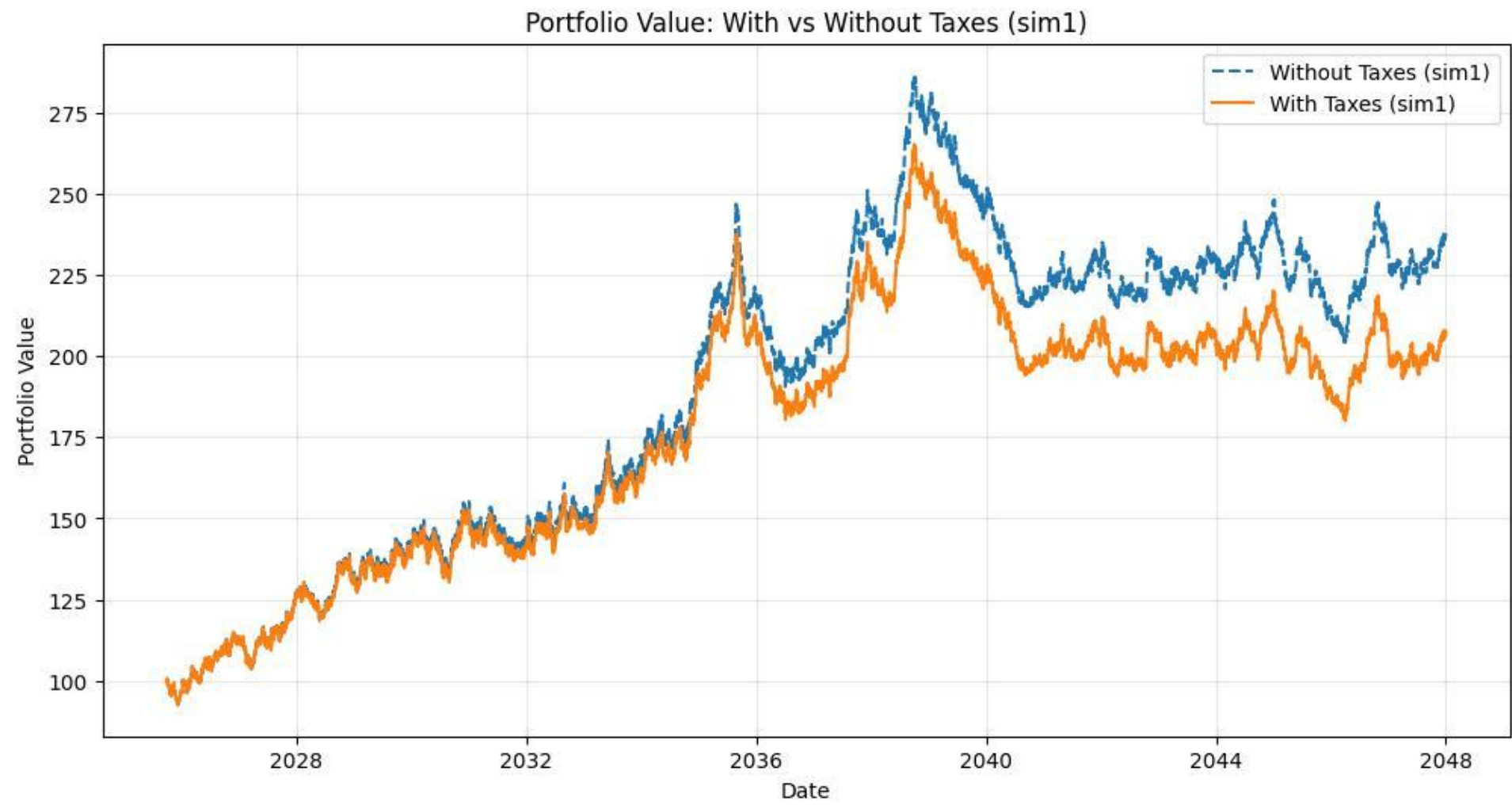
Glide Path Allocation by Positions



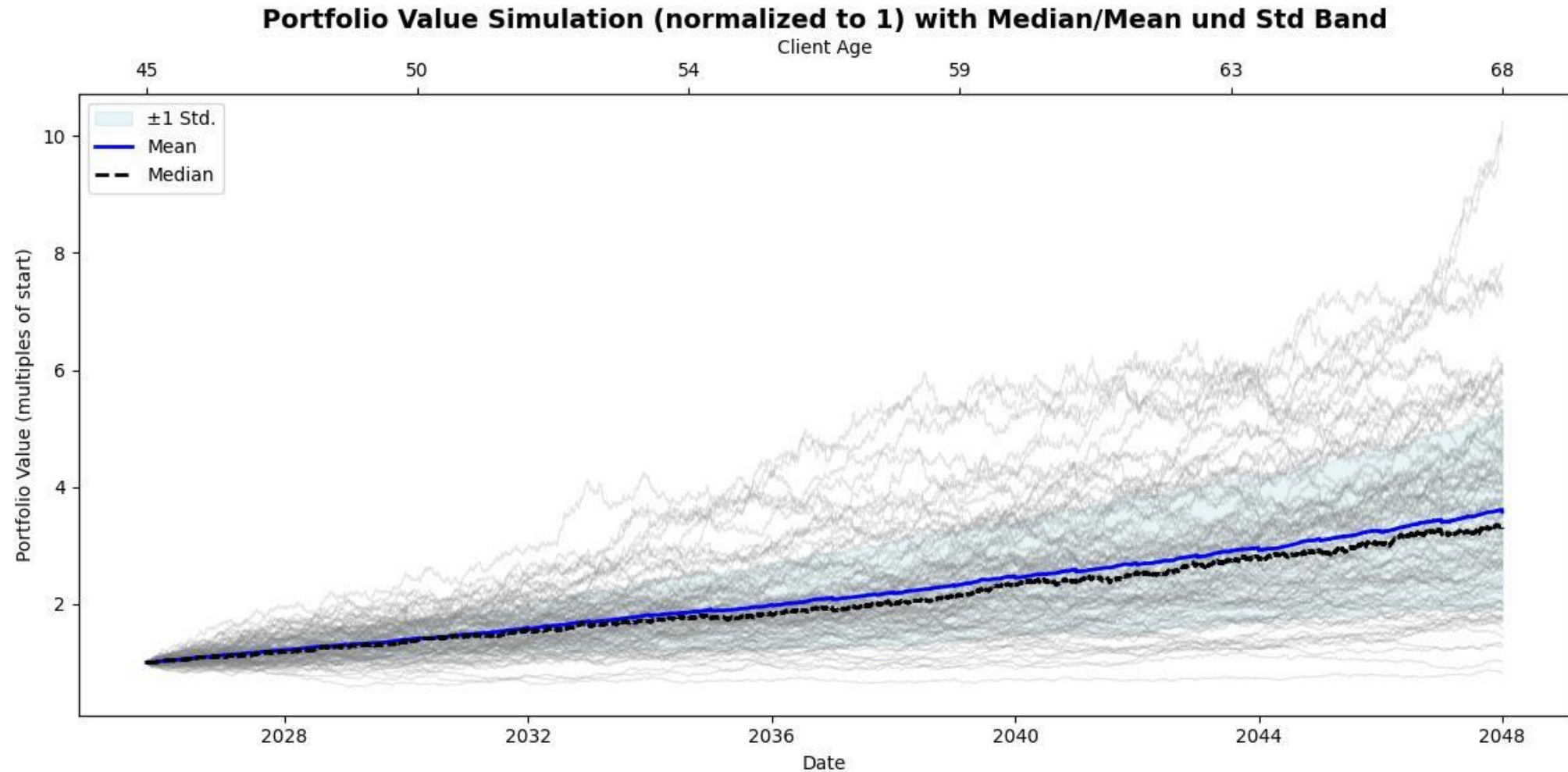
Glide Path Allocation by Positions



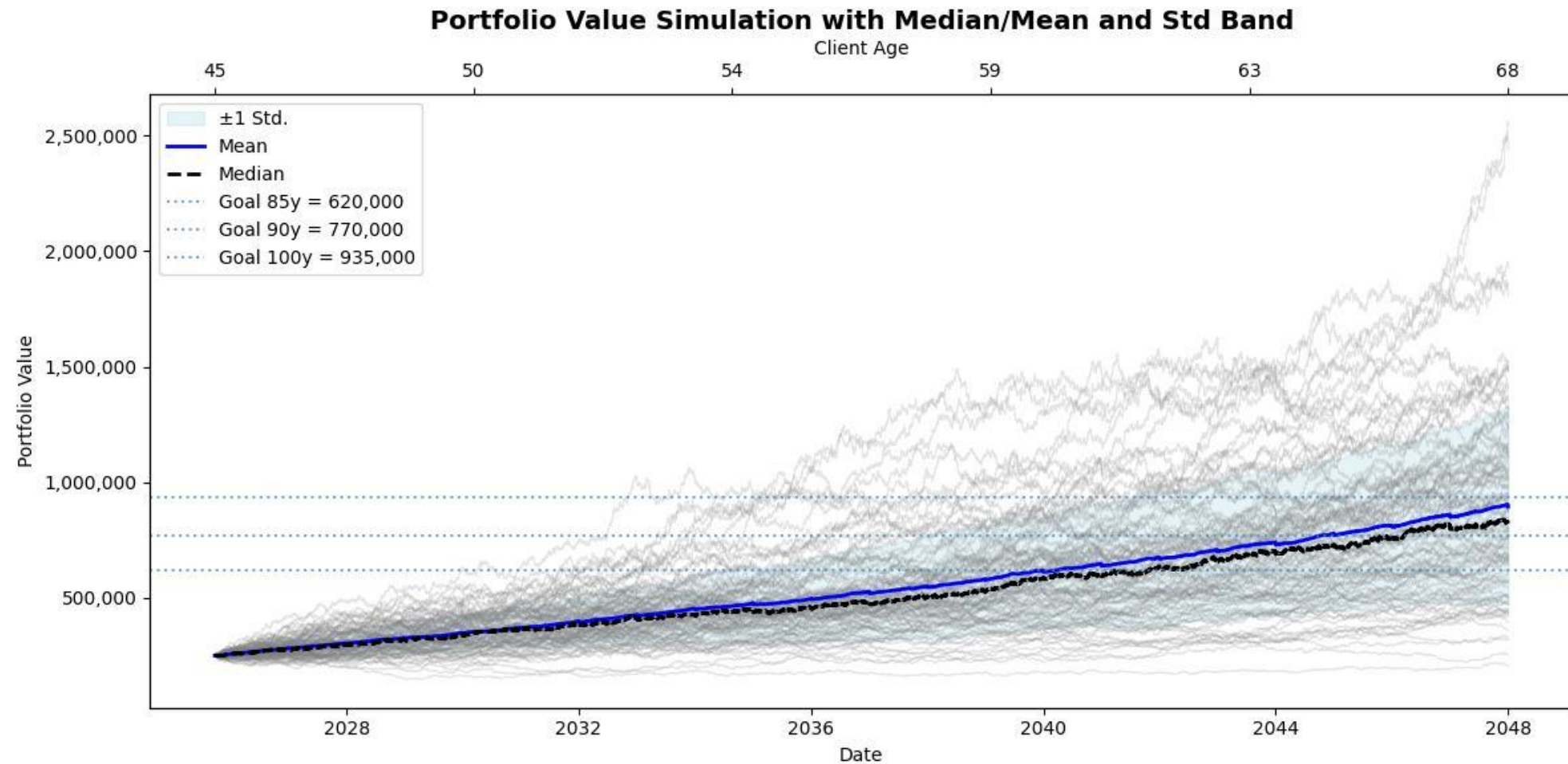
Impact of Taxes on Portfolio Value



Rebalancing with Monte Carlo Simulation of Portfolio Value



Rebalancing with Monte Carlo Simulation of Portfolio Value



Retirement Investment Risks

“We have a portfolio... but what could derail it?”

Even a well-constructed portfolio faces **external risks** that can threaten retirement security:

- **Longevity Risk** → living longer than expected, exhausting assets.
- **Inflation Risk** → erosion of purchasing power over decades due to high inflation.
- **Market Risk** → Economic downturns near retirement causing permanent losses.
- **Withdrawal Risk** → unsustainable drawdowns accelerating depletion.
- **Tax and Policy risk** → Policy changes causing the expected return to drop due to increased costs/taxes.

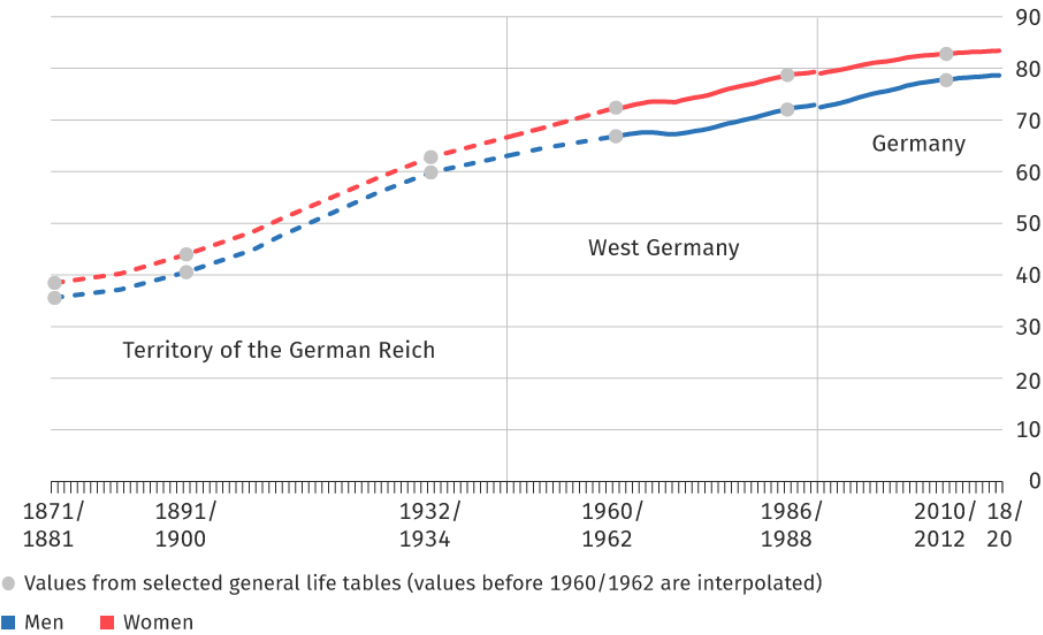
To be well informed and make our portfolio resilient, we must now examine these risks and the strategies we could use to mitigate them.

Longevity Risk - Context

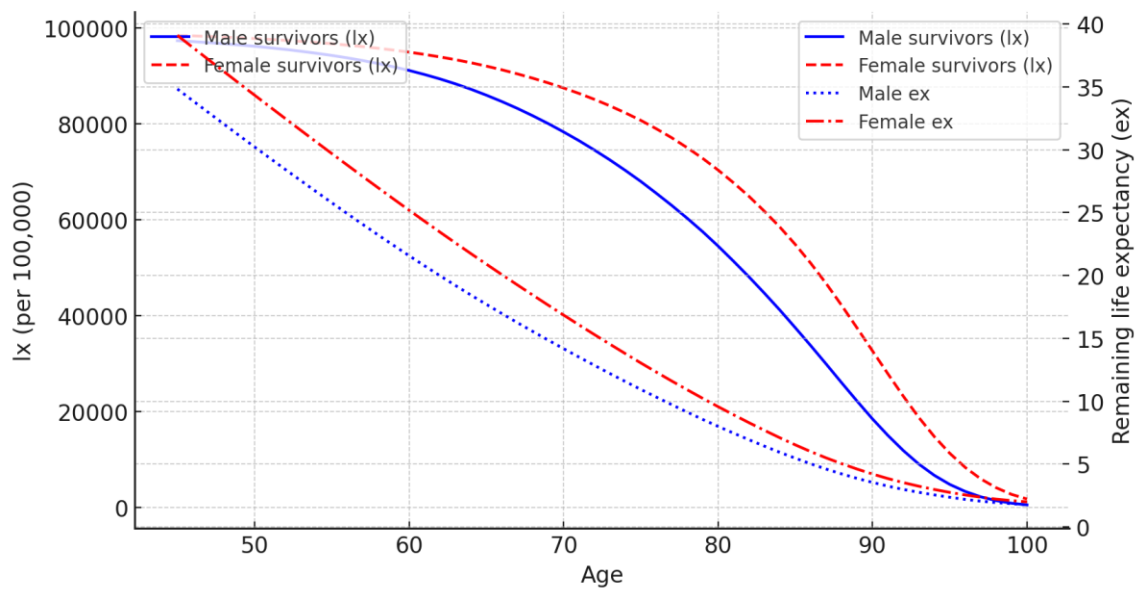
A longer-than-expected lifespan may significantly strain retirement resources, potentially exhausting the investment portfolio.

Life expectancy at birth

Years



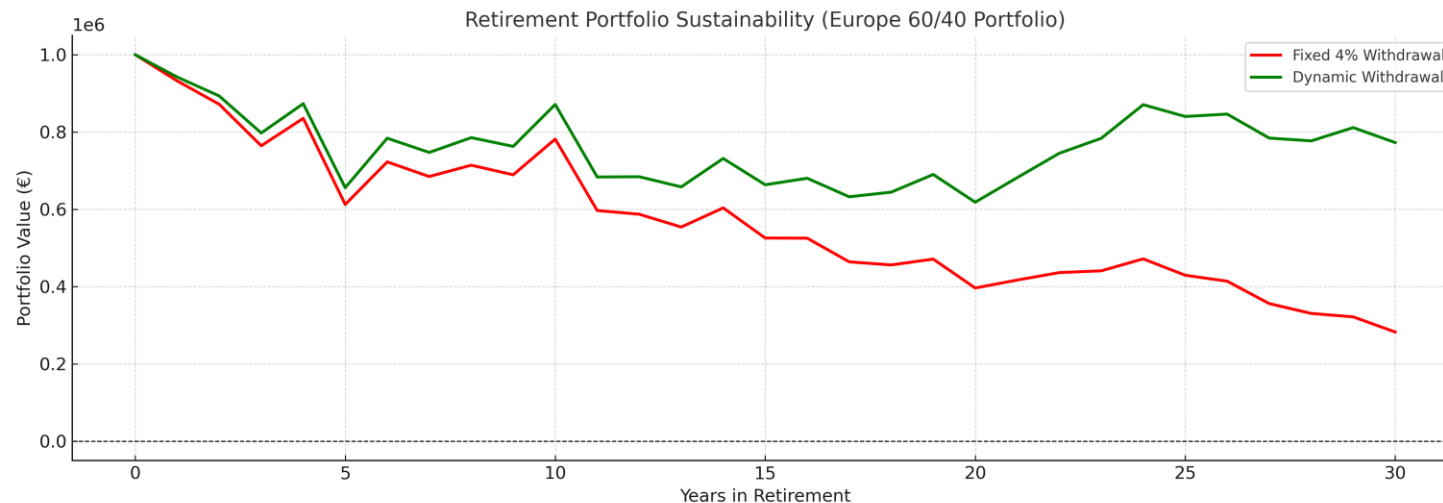
Germany Life Table 2022/2024 (Age 45-100) - Male vs Female



Longevity Risk - Mitigation

Longevity risk can be mitigated through withdrawal strategies and maintaining growth potential throughout retirement.

- ▶ **Maintain equity exposure in retirement (20–30%)** : Continuing equity allocation supports growth over long horizons, hedging against outliving assets.
- ▶ **Dynamic withdrawal rules (flexible spending)** : Adjust withdrawals up or down depending on market returns to avoid depleting assets.

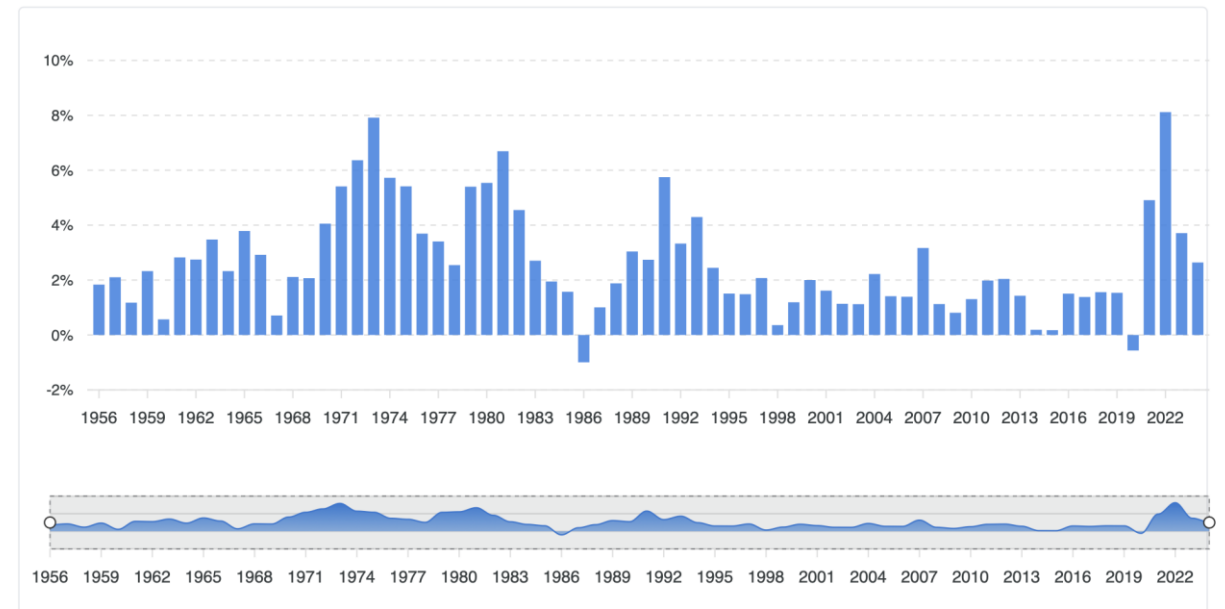


Inflation Risk - Context

Inflation erodes the purchasing power of retirement income. Even modest inflation compounds significantly over 20–30 years.

- ▶ **1990s-2010s: low and stable inflation (1- 2%)**
- ▶ **Inflation rate spiked to 8% due to the energy crisis and supply chain shocks** : Retirees cannot rely on “low inflation” assumptions.

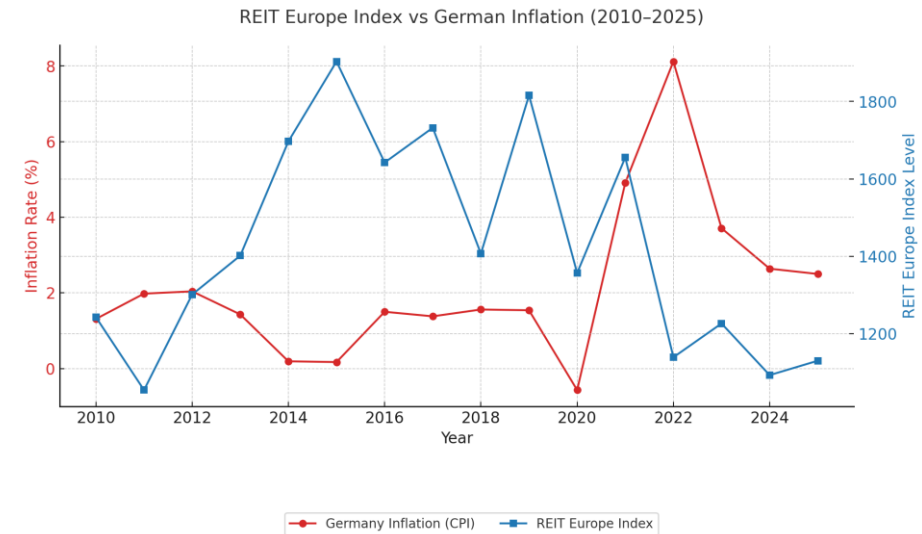
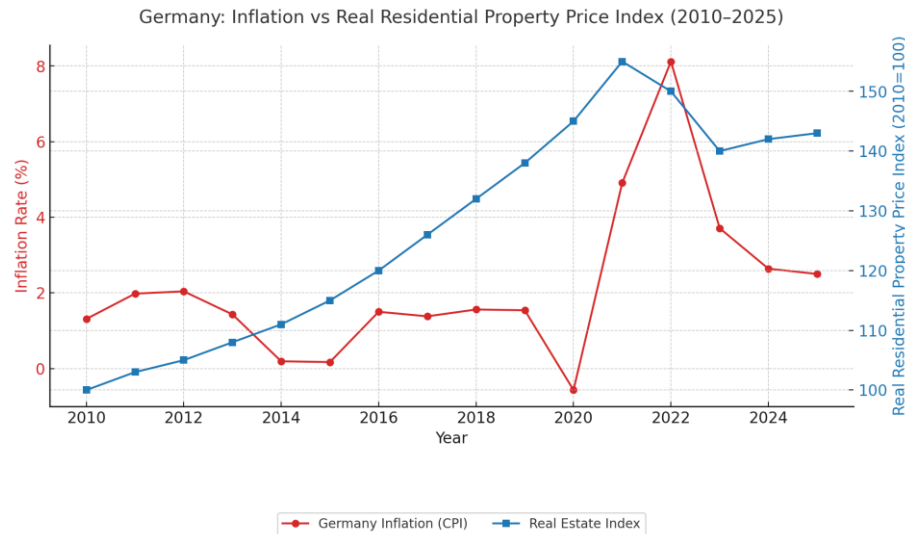
Germany Historical Inflation Rates



Inflation Risk - Mitigation

To protect retirement income from inflation, portfolios must include strategies that preserve real purchasing power across different economic environments.

- ▶ **Inflation-Linked Bonds (ILBs/TIPS)** : Euro ILBs provide principal and coupon indexed to inflation → stable real income stream.
- ▶ **Real Estate / REITs** : Property values and rents tend to rise with inflation, providing a partial hedge.



Market Risk – Context & Mitigation

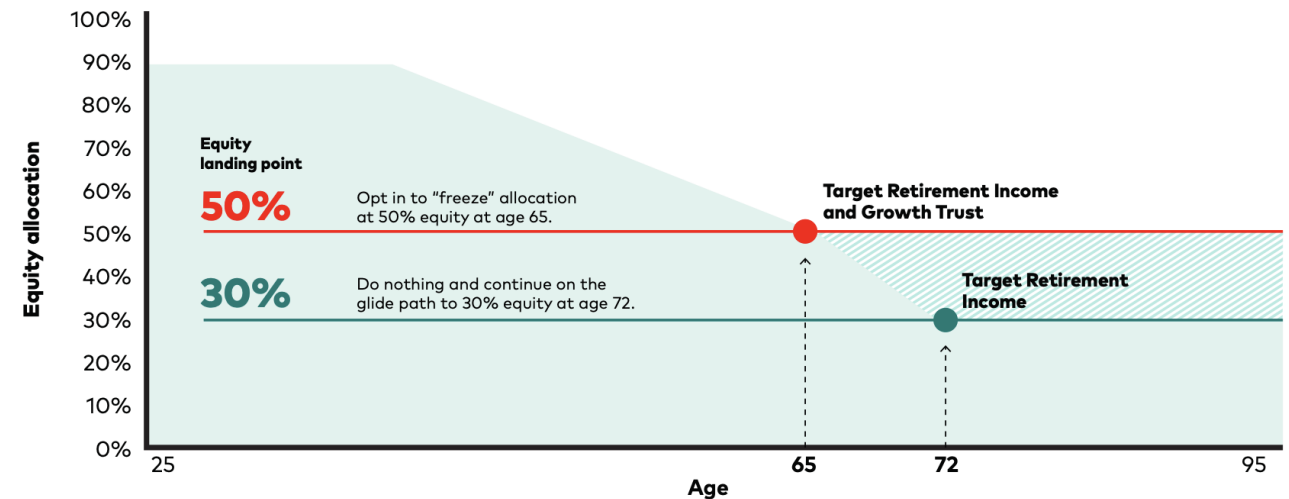
If such a crash occurs close to retirement age, the portfolio may not recover, leading to a permanent shortfall.

► **Example :** In 2008, global equities fell around 40%, so a investor with a 60/40 portfolio could have lost about a quarter of the wealth in just one year.

► **Mitigation:** Glide path reduces equity exposure before retirement.

> eg. Investors may choose different “landing points,” but industry standard is to de-risk near retirement.

Equity allocation of participants who self-select into a higher-equity landing option



Source: Vanguard.

Notes: TRIGT is available only as a collective investment trust. Target Retirement Income is available as both a trust and a 1940 Act mutual fund.

Tax Risk – Context & Mitigation

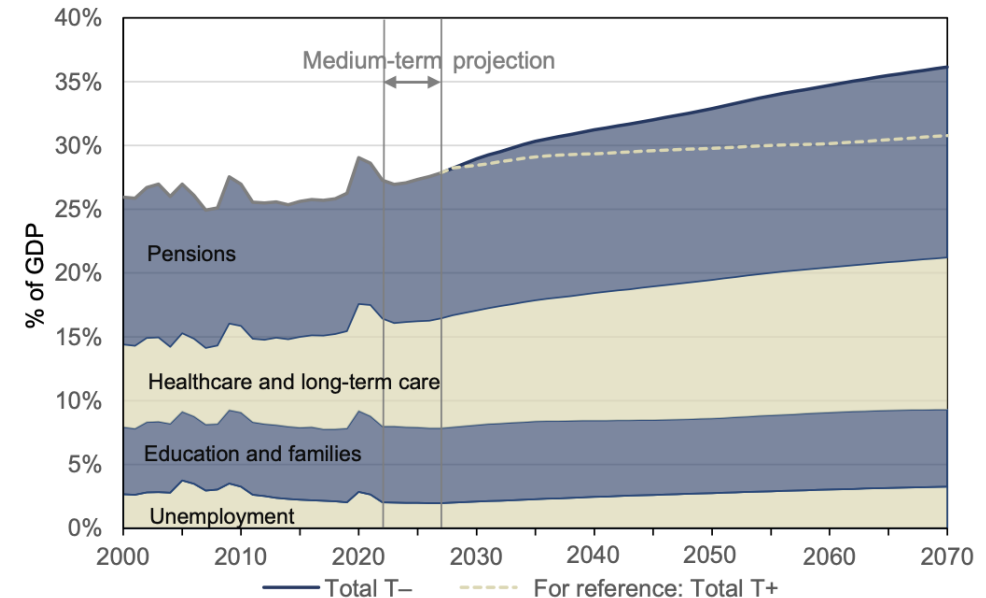
Future tax and policy changes can reduce net retirement income, so projections must include buffers and tax diversification.

► **Impressions** : German capital income currently taxed at flat 25%. But with policy uncertainty, retirement projections may overstate future net income.

► **Mitigation:** Build a Buffer in Projections.
> Incorporate a haircut on projected net returns (e.g., assume future taxation rises by 5-10%).

Demographically sensitive public expenditure (2000-2070), variant T-

Figure 9



Team



Ying Liu
8513629



Abdul Malik
8512905



Marco Leibersperger
8518121



Ritesh Kumar
8510249



Shangrun Jiang
8501084



Mujtaba Ali Bhutto
8513611



Chang Liu
8514049

Thank you !

APENDIX

Salary Projections and Tax

	Salary and tax calculations												
Year	Age	Bruotto	monthly brutto	credit earned	Cumulative EP	pension tax rate (50%)	Arbeitslosenversicherung	Pflegeversicherung	medical insurance	zvE_m	zvE_y	zvE_m_tax	netto
2010	30	36000	3000	1.16	1.16	279.00	39.00	51.00	350.00	2281	27372	55	2277
2011	31	38400	3200	1.20	2.35	297.60	41.60	54.40	350.00	2456	29477	88	2423
2012	32	40800	3400	1.24	3.59	316.20	44.20	57.80	350.00	2632	31582	125	2565
2013	33	43200	3600	1.28	4.87	334.80	46.80	61.20	350.00	2807	33686	164	2704
2014	34	45600	3800	1.32	6.19	353.40	49.40	64.60	350.00	2983	35791	197	2850
2015	35	48000	4000	1.36	7.55	372.00	52.00	68.00	350.00	3158	37896	240	2986
2016	36	52800	4400	1.46	9.01	409.20	57.20	74.80	350.00	3509	42106	328	3256
2017	37	57600	4800	1.55	10.56	446.40	62.40	81.60	350.00	3860	46315	418	3523
2018	38	62400	5200	1.63	12.20	483.60	67.60	88.40	350.00	4210	50525	512	3787
2019	39	67200	5600	1.76	13.95	520.80	72.80	95.20	350.00	4561	54734	608	4049
2020	40	72000	6000	1.83	15.79	558.00	78.00	102.00	350.00	4912	58944	706	4308
2021	41	73920	6160	1.89	17.67	572.88	80.08	104.72	350.00	5052	60628	746	4411
2022	42	75840	6320	1.87	19.55	587.76	82.16	107.44	350.00	5193	62312	787	4513
2023	43	77760	6480	1.92	21.47	602.64	84.24	110.16	350.00	5333	63996	828	4615
2024	44	79680	6640	1.89	23.36	617.52	86.32	112.88	350.00	5473	65679	869	4717
2025	45	81600	6800	1.62	24.98	632.40	88.40	115.60	350.00	5614	67363	911	4818
2026	46	83232	6936	1.62	26.60	645.05	90.17	117.91	360.50	5722	68668	944	4896
2027	47	84897	7075	1.62	28.21	657.95	91.97	120.27	360.50	5844	70128	981	4983
2028	48	86595	7216	1.62	29.83	671.11	93.81	122.68	360.50	5968	71617	1019	5072
2029	49	88326	7361	1.62	31.45	684.53	95.69	125.13	360.50	6095	73136	1058	5162
2030	50	90093	7508	1.62	33.06	698.22	97.60	127.63	371.32	6213	74556	1095	5245
2031	51	91895	7658	1.62	34.68	712.19	99.55	130.18	371.32	6345	76136	1137	5338
2032	52	93733	7811	1.62	36.29	726.43	101.54	132.79	371.32	6479	77748	1179	5433
2033	53	95607	7967	1.62	37.91	740.96	103.57	135.44	371.32	6616	79392	1223	5528
2034	54	97520	8127	1.62	39.53	755.78	105.65	138.15	382.45	6745	80935	1265	5618
2035	55	99470	8289	1.62	41.14	770.89	107.76	140.92	382.45	6887	82646	1311	5717
2036	56	101459	8455	1.62	42.76	786.31	109.91	143.73	352.45	7063	84750	1369	5837
2037	57	103489	8624	1.62	44.37	802.04	112.11	146.61	352.45	7211	86530	1418	5939
2038	58	105558	8797	1.62	45.99	818.08	114.35	149.54	363.03	7352	88218	1466	6035
2039	59	107669	8972	1.62	47.61	834.44	116.64	152.53	333.03	7536	90430	1528	6160

Pension EP Projections

Year	Age	Inflation Rate	Durchschnittsentgelt increase rate	Durchschnittsentgelt	PKV increase rate	PKV	1 Credit	pension tax rate (50%)	Arbeitslosenversicherung (50%)	Pflegeversicherung (50%)	Kindergeld	Kid#
2010	30			31144		700.00		18.60%	2.60%	3.40%	250	0
2011	31		3.1%	32100		700.00		18.60%	2.60%	3.40%	250	0
2012	32		2.8%	33002		700.00		18.60%	2.60%	3.40%	250	0
2013	33		2.0%	33659		700.00		18.60%	2.60%	3.40%	250	0
2014	34		2.5%	34514		700.00		18.60%	2.60%	3.40%	250	0
2015	35		2.5%	35363		700.00		18.60%	2.60%	3.40%	250	0
2016	36		2.3%	36187		700.00		18.60%	2.60%	3.40%	250	1
2017	37		2.5%	37077		700.00		18.60%	2.60%	3.40%	250	1
2018	38		3.1%	38212		700.00		18.60%	2.60%	3.40%	250	1
2019	39		0.0%	38212		700.00		18.60%	2.60%	3.40%	250	2
2020	40		2.8%	39301		700.00		18.60%	2.60%	3.40%	250	2
2021	41		-0.3%	39167		700.00		18.60%	2.60%	3.40%	250	2
2022	42		3.3%	40463		700.00		18.60%	2.60%	3.40%	250	2
2023	43		0.0%	40463		700.00		18.60%	2.60%	3.40%	250	2
2024	44		3.9%	42053		700.00		18.60%	2.60%	3.40%	250	2
2025	45	2.00%	20.1%	50493.00		700.00	40.79	18.60%	2.60%	3.40%	250	2
2026	46	2.00%	2.00%	51502.86	3.00%	721.00	40.79	18.60%	2.60%	3.40%	250	2
2027	47	2.00%	2.00%	52532.92		721.00	40.79	18.60%	2.60%	3.40%	250	2
2028	48	2.00%	2.00%	53583.58		721.00	40.79	18.60%	2.60%	3.40%	250	2
2029	49	2.00%	2.00%	54655.25		721.00	40.79	18.60%	2.60%	3.40%	250	2
2030	50	2.00%	2.00%	55748.35	3.00%	742.63	40.79	18.60%	2.60%	3.40%	250	2
2031	51	2.00%	2.00%	56863.32		742.63	40.79	18.60%	2.60%	3.40%	250	2
2032	52	2.00%	2.00%	58000.59		742.63	40.79	18.60%	2.60%	3.40%	250	2
2033	53	2.00%	2.00%	59160.60		742.63	40.79	18.60%	2.60%	3.40%	250	2
2034	54	2.00%	2.00%	60343.81	3.00%	764.91	40.79	18.60%	2.60%	3.40%	250	2
2035	55	2.00%	2.00%	61550.69		764.91	40.79	18.60%	2.60%	3.40%	250	2
2036	56	2.00%	2.00%	62781.70		704.91	40.79	18.60%	2.60%	3.40%	250	1
2037	57	2.00%	2.00%	64037.33		704.91	40.79	18.60%	2.60%	3.40%	250	1
2038	58	2.00%	2.00%	65318.08	3.00%	726.06	40.79	18.60%	2.60%	3.40%	250	1
2039	59	2.00%	2.00%	66624.44		666.06	40.79	18.60%	2.60%	3.40%	250	0
2040	60	2.00%	2.00%	67956.93		666.06	40.79	18.60%	2.60%	3.40%	250	0

Pension Projections and Tax

Year	Age	Bruotto	monthly brutto	medical insurance	zvE_m	zvE_y	zvE_m_tax	netto
2048	68	46,724	3,894	707	3,012	36,142	204	3,690
2049	69	46,724	3,972	707	3,086	37,034	222	3,749
2050	70	47,659	4,051	728	3,141	37,690	236	3,815
2051	71	47,659	4,132	728	3,218	38,619	255	3,877
2052	72	49,584	4,215	728	3,297	39,566	274	3,940
2053	73	49,584	4,299	728	3,378	40,532	295	4,004
2054	74	51,587	4,385	750	3,438	41,255	310	4,075
2055	75	51,587	4,473	750	3,522	42,260	331	4,142
2056	76	53,671	4,562	750	3,607	43,285	353	4,209
2057	77	53,671	4,653	750	3,694	44,331	375	4,278
2058	78	55,840	4,746	772	3,761	45,128	393	4,354
2059	79	55,840	4,841	772	3,851	46,216	416	4,425
2060	80	58,096	4,938	772	3,944	47,325	440	4,498
2061	81	58,096	5,037	772	4,038	48,457	465	4,571
2062	82	60,443	5,138	795	4,111	49,333	485	4,653
2063	83	60,443	5,240	795	4,209	50,511	511	4,729
2064	84	62,884	5,345	795	4,309	51,712	538	4,807
2065	85	62,884	5,452	795	4,411	52,937	566	4,886
2066	86	65,425	5,561	819	4,492	53,900	588	4,973
2067	87	65,425	5,672	819	4,598	55,175	618	5,055
2068	88	68,068	5,786	819	4,706	56,475	648	5,138
2069	89	68,068	5,902	819	4,817	57,801	679	5,222
2070	90	70,818	6,020	844	4,905	58,859	704	5,315
2071	91	70,818	6,140	844	5,020	60,239	737	5,403
2072	92	73,679	6,263	844	5,137	61,646	771	5,492
2073	93	73,679	6,388	844	5,257	63,081	806	5,582
2074	94	76,656	6,516	869	5,353	64,242	834	5,682
2075	95	76,656	6,646	869	5,478	65,735	871	5,775
2076	96	79,753	6,779	869	5,605	67,258	909	5,870

Cashflow Projections

Year	Age	CF Need	Netto Pension Contribution	Net Cash Needed from Investment
2047	67	-	-	0
2048	68	69,666	29,630	40,035
2049	69	71,766	29,630	42,135
2050	70	73,929	29,630	44,298
2051	71	76,135	29,630	46,505
2052	72	78,386	29,630	48,755
2053	73	80,681	29,630	51,051
2054	74	83,044	29,630	53,414
2055	75	85,455	29,630	55,825
2056	76	87,914	29,630	58,283
2057	77	90,422	29,630	60,791
2058	78	93,002	29,630	63,372
2059	79	95,634	29,630	66,004
2060	80	98,319	29,630	68,689
2061	81	101,058	29,630	71,427
2062	82	103,874	29,630	74,244

Year	Age	CF Need	Netto Pension Contribution	Net Cash Needed from Investment
2063	83	106,747	29,630	77,117
2064	84	109,677	29,630	80,047
2065	85	112,666	29,630	83,036
2066	86	115,739	29,630	86,108
2067	87	118,873	29,630	89,242
2068	88	122,069	29,630	92,439
2069	89	125,330	29,630	95,699
2070	90	128,680	29,630	99,050
2071	91	132,097	29,630	102,467
2072	92	135,583	29,630	105,953
2073	93	139,138	29,630	109,508
2074	94	142,790	29,630	113,160
2075	95	146,515	29,630	116,885
2076	96	150,315	29,630	120,684
2077	97	154,190	29,630	124,560
2078	98	158,169	29,630	128,538
2079	99	162,227	29,630	132,597
2080	100	166,367	29,630	136,737